

A TECHNICAL NOTE ON SIGNATURE COMPUTATIONS WITH BIDEGREE TRUNCATION, COORDINATE CHANGES, AND SYMMETRIZATION

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ABSTRACT. We give explicit data layouts and algorithms for signature computations with bidegree truncation, the coordinate transformation introduced in [2], and symmetrization over a partial alphabet. The main conceptual difficulty is providing a contiguous array layout for the bidegree-truncated tensors and expressing the basic tensor-algebra operations, concatenation and shuffle, efficiently within this layout. Our main approach is to introduce a rank axis that accounts for the colexicographic ordering of the positions of the letters of the first alphabet. All operations can then be expressed as operations on the rank axis and as permutations of the remaining dense tensor axes corresponding to the partial alphabets. We provide further implementation details for vectorized array execution and discuss how word-parallel signature computation interfaces with the ordered and partially symmetrized bidegree layouts.

Disclaimer. To distinguish this paper from our other publications, we note that every part of the manuscript was written and revised through iterative human-machine cycles of prompting, generation, and review using the large language models *GBT-5.6 Sol* and *Claude Fabel 5.0*. All algorithms proposed in the paper were implemented in `tensordev`, and both the implementations and the examples were verified through an extensive suite of unit tests. The authors reviewed and approved all generated material and take full responsibility for the final content.

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1. ALGEBRAIC PRELIMINARIES

1.1. Split alphabets, words, and bidegrees. Let $\mathcal{A} = \mathcal{A}' \dot{\cup} \mathcal{A}''$ be a finite alphabet with both $\mathcal{A}'$ and $\mathcal{A}''$ nonempty, and define $d := |\mathcal{A}|$. We write $i \in \mathcal{A}'$ and $j \in \mathcal{A}''$, and denote by $\mathcal{W}_{\mathcal{A}}$ the set of finite words over $\mathcal{A}$, including the empty word $\emptyset$. For $w, v \in \mathcal{W}_{\mathcal{A}}$, we write $wv \in \mathcal{W}_{\mathcal{A}}$ for their concatenation.

For $w \in \mathcal{W}_{\mathcal{A}}$, let $|w|_{\mathcal{A}'}$ and $|w|_{\mathcal{A}''}$ be the numbers of letters of w in $\mathcal{A}'$ and $\mathcal{A}''$, respectively. We call $(|w|_{\mathcal{A}'}, |w|_{\mathcal{A}''}) \in \mathbb{N}_0 \times \mathbb{N}_0$ the *bidegree* of w . In particular, $|w| = |w|_{\mathcal{A}'} + |w|_{\mathcal{A}''}$. Every word w of bidegree (n, m) has a unique decomposition

$$w = w_0 i_1 w_1 \cdots i_n w_n, \quad w_0, \dots, w_n \in \mathcal{W}_{\mathcal{A}'}, \quad i_1, \dots, i_n \in \mathcal{A}',$$

with $|w_0| + \cdots + |w_n| = m$. We write

$$\Delta_{\mathcal{A}'}(w) := ((w_0, \dots, w_n), i_1 \cdots i_n).$$

For $n = 0$, this means $\Delta_{\mathcal{A}'}(w) = ((w), \emptyset)$.

A path $Y: [s, t] \rightarrow \mathbb{R}^d$ whose coordinates are indexed by $\mathcal{A}$ is written as $Y = (A, X)$, where A consists of the coordinates indexed by $\mathcal{A}'$ and X of those indexed by $\mathcal{A}''$.

1.2. Tensor algebra, signatures and shuffle. Let $(e_i)_{i \in \mathcal{A}}$ be the canonical basis of $\mathbb{R}^d$. For $w = i_1 \cdots i_r \in \mathcal{W}_{\mathcal{A}}$, write $e_w := e_{i_1} \otimes \cdots \otimes e_{i_r}$ and $e_{\emptyset} := 1$. The extended tensor algebra is

$$T((\mathbb{R}^d)) := \prod_{r \geq 0} (\mathbb{R}^d)^{\otimes r}, \quad (\mathbb{R}^d)^{\otimes 0} := \mathbb{R},$$

with concatenation product $\otimes$, determined by $e_u \otimes e_v = e_{uv}$.

The free associative algebra $\mathbb{R}\langle \mathcal{A} \rangle$ is the linear span of $\mathcal{W}_{\mathcal{A}}$. We identify a word with the corresponding coordinate functional on $T((\mathbb{R}^d))$ and write $\langle w, \mathbf{a} \rangle$ for the coefficient of e_w in $\mathbf{a}$. This extends to the bilinear pairing

$$\langle \cdot, \cdot \rangle: \mathbb{R}\langle \mathcal{A} \rangle \times T((\mathbb{R}^d)) \longrightarrow \mathbb{R}.$$

Let $Y \in C^{1\text{-var}}([s, t], \mathbb{R}^d)$. For $w = i_1 \cdots i_n$, its signature coordinate is

$$\langle w, \text{Sig}(Y)_{s,t} \rangle := \int_{s < u_1 < \cdots < u_n < t} dY_{u_1}^{i_1} \cdots dY_{u_n}^{i_n}, \quad \langle \emptyset, \text{Sig}(Y)_{s,t} \rangle := 1.$$

The signature is the tensor series $\text{Sig}(Y)_{s,t} = \sum_{w \in \mathcal{W}_{\mathcal{A}}} \langle w, \text{Sig}(Y)_{s,t} \rangle e_w$. For $s \leq u \leq t$, Chen's identity states that

$$\text{Sig}(Y)_{s,t} = \text{Sig}(Y)_{s,u} \otimes \text{Sig}(Y)_{u,t}.$$

If $Y_{s,\cdot}$ is linear on $[s, t]$, then

$$\text{Sig}(Y)_{s,t} = \exp_{\otimes}(Y_{s,t}) := \sum_{r=0}^{\infty} \frac{Y_{s,t}^{\otimes r}}{r!}.$$

Consequently, the signature of a piecewise linear path is obtained by tensor exponentiation of each increment and combining the segment signatures by concatenation.

The shuffle product on $\mathbb{R}\langle \mathcal{A} \rangle$ is determined recursively by

$$w \sqcup \emptyset = \emptyset \sqcup w = w, \quad wi \sqcup vj = (w \sqcup vj)i + (wi \sqcup v)j,$$

for $w, v \in \mathcal{W}_{\mathcal{A}}$ and $i, j \in \mathcal{A}$. Signature coordinates satisfy the shuffle identity

$$\langle u, \text{Sig}(Y)_{s,t} \rangle \langle v, \text{Sig}(Y)_{s,t} \rangle = \langle u \sqcup v, \text{Sig}(Y)_{s,t} \rangle.$$

1.3. Bidegree truncation. For the alphabet split $\mathcal{A} = \mathcal{A}' \dot{\cup} \mathcal{A}''$ and $n, m \in \mathbb{N}_0$, define the homogeneous bidegree spaces

$$\begin{aligned} T^{(n,m)}(\mathbb{R}^d) &:= \text{span} \{e_w : |w|_{\mathcal{A}'} = n, \quad |w|_{\mathcal{A}''} = m\}, \\ \mathbb{R}^{(n,m)}\langle \mathcal{A} \rangle &:= \text{span} \{w : |w|_{\mathcal{A}'} = n, \quad |w|_{\mathcal{A}''} = m\}. \end{aligned}$$

For $\mathbf{a} \in T((\mathbb{R}^d))$, we write $\mathbf{a}^{(n,m)} \in T^{(n,m)}(\mathbb{R}^d)$ for its homogeneous component of bidegree (n, m) , and analogously for elements of $\mathbb{R}\langle \mathcal{A} \rangle$. For truncation levels $N, M \in \mathbb{N}_0$, define the bigraded truncated tensor algebra and dual word space by

$$T^{N,M}(\mathbb{R}^d) := \bigoplus_{n=0}^N \bigoplus_{m=0}^M T^{(n,m)}(\mathbb{R}^d), \quad \mathbb{R}^{N,M}\langle \mathcal{A} \rangle := \bigoplus_{n=0}^N \bigoplus_{m=0}^M \mathbb{R}^{(n,m)}\langle \mathcal{A} \rangle.$$

We denote the corresponding projections by $\pi^{N,M}$, so that

$$\pi^{N,M} \mathbf{a} = \sum_{n=0}^N \sum_{m=0}^M \mathbf{a}^{(n,m)},$$

and use the same notation for the projection on $\mathbb{R}\langle \mathcal{A} \rangle$. Since concatenation and shuffle add bidegrees, they descend to the bigraded truncation through

$$\mathbf{a} \otimes_{N,M} \mathbf{b} := \pi^{N,M}(\mathbf{a} \otimes \mathbf{b}), \quad \alpha \sqcup_{N,M} \beta := \pi^{N,M}(\alpha \sqcup \beta).$$

Both truncated products are associative and have the empty word as their unit.

We write $\text{Sig}(Y)_{s,t}^{N,M} := \pi^{N,M} \text{Sig}(Y)_{s,t}$. Since the discarded bidegrees form an ideal for concatenation, Chen's identity descends to

$$\text{Sig}(Y)_{s,t}^{N,M} = \text{Sig}(Y)_{s,u}^{N,M} \otimes_{N,M} \text{Sig}(Y)_{u,t}^{N,M}.$$

Moreover, for words u, v satisfying $|u|_{\mathcal{A}'} + |v|_{\mathcal{A}'} \leq N$ and $|u|_{\mathcal{A}''} + |v|_{\mathcal{A}''} \leq M$, the shuffle identity becomes

$$\langle u, \text{Sig}(Y)_{s,t}^{N,M} \rangle \langle v, \text{Sig}(Y)_{s,t}^{N,M} \rangle = \langle u \sqcup v, \text{Sig}(Y)_{s,t}^{N,M} \rangle.$$

For $K \in \mathbb{N}_0$, the total-degree truncated tensor algebra of degree K is

$$T^K(\mathbb{R}^d) := \bigoplus_{r=0}^K (\mathbb{R}^d)^{\otimes r}, \quad \pi_K : T((\mathbb{R}^d)) \rightarrow T^K(\mathbb{R}^d), \quad \mathbf{a} \otimes_K \mathbf{b} := \pi_K(\mathbf{a} \otimes \mathbf{b}).$$

For $K = N + M$, it is the smallest total-degree truncated tensor algebra containing $T^{N,M}(\mathbb{R}^d)$.

1.4. Shear coordinate transformation. We recall the coordinate transformation associated with the alphabet split $(\mathcal{A}', \mathcal{A}'')$ introduced in [2], which we call the *shear coordinate transformation* for simplicity; cf. [2, Remark 3.3]. Specifically, the transformation is given by a graded linear automorphism $\Psi : T((\mathbb{R}^d)) \rightarrow T((\mathbb{R}^d))$ and the induced pairing on $\mathbb{R}\langle \mathcal{A} \rangle \times T((\mathbb{R}^d))$ defined by

$$\langle \alpha, \mathbf{a} \rangle_\Psi := \langle \alpha, \Psi(\mathbf{a}) \rangle, \quad \alpha \in \mathbb{R}\langle \mathcal{A} \rangle, \quad \mathbf{a} \in T((\mathbb{R}^d)).$$

The transformation is characterized by the signature coefficients in the resulting coordinates $\langle \cdot, \cdot \rangle_\Psi$. More precisely, let $Y = (A, X) \in C^{1\text{-var}}([s, t], \mathbb{R}^d)$. For $w \in \mathcal{W}_{\mathcal{A}}$, write $\Delta_{\mathcal{A}'}(w) = ((w_0, \dots, w_n), \mathbf{i}_1 \cdots \mathbf{i}_n)$. Set $\Delta_{s,t}^n := \{(t_1, \dots, t_n) : s < t_1 < \dots < t_n < t\}$, with the usual empty-simplex convention for $n = 0$. Then the corresponding signature coordinate is

$$\langle w, \text{Sig}(Y)_{s,t} \rangle_\Psi = \langle w_n, \text{Sig}(X)_{s,t} \rangle \int_{\Delta_{s,t}^n} \prod_{j=1}^n \langle w_{j-1}, \text{Sig}(X)_{s,t_j} \rangle dA_{t_j}^{i_j}.$$

Since Ψ is an invertible graded linear map, its inverse $\Psi^{-1} : T((\mathbb{R}^d)) \rightarrow T((\mathbb{R}^d))$ is graded as well. We recall the recursive algebraic definitions of both maps in [Section 4](#). Moreover, both maps preserve bidegrees, that is, for every $\mathbf{a} \in T((\mathbb{R}^d))$ and $(n, m) \in \mathbb{N}_0^2$,

$$(\Psi(\mathbf{a}))^{(n,m)} = \Psi(\mathbf{a}^{(n,m)}), \quad (\Psi^{-1}(\mathbf{a}))^{(n,m)} = \Psi^{-1}(\mathbf{a}^{(n,m)}).$$

Consequently, they restrict to inverse linear automorphisms of $T^{N,M}(\mathbb{R}^d)$.

1.5. Partial symmetrization and quotient normal forms. Following [\[2\]](#), we partially symmetrize by imposing commutativity between adjacent letters in $\mathcal{A}''$. Let

$$\mathcal{J}_{\mathcal{A}''} := \text{span} \{ u(\textcolor{brown}{i}\textcolor{brown}{j} - \textcolor{brown}{j}\textcolor{brown}{i})v : u, v \in \mathcal{W}_{\mathcal{A}}, \quad \textcolor{brown}{i}, \textcolor{brown}{j} \in \mathcal{A}'' \}, \quad \mathbb{R}_{\mathcal{A}''} \langle \mathcal{A} \rangle := \mathbb{R} \langle \mathcal{A} \rangle / \mathcal{J}_{\mathcal{A}''},$$

and denote the quotient class of a word w by $[w]$. If $\Delta_{\mathcal{A}'}(w) = ((w_0, \dots, w_n), \textcolor{blue}{i}_1 \cdots \textcolor{blue}{i}_n)$, then

$$[w] = [w_0]\textcolor{blue}{i}_1[w_1] \cdots \textcolor{blue}{i}_n[w_n].$$

Thus only permutations within an $\mathcal{A}''$ -block are identified, while the order of the blocks and of the $\mathcal{A}'$ -letters is retained. The ideal $\mathcal{J}_{\mathcal{A}''}$ is bigraded, and the quotient therefore inherits the bidegree decomposition. For $n, m \in \mathbb{N}_0$, define

$$\mathbb{R}_{\mathcal{A}''}^{(n,m)} \langle \mathcal{A} \rangle := \text{span} \{ [w] : |w|_{\mathcal{A}'} = n, \quad |w|_{\mathcal{A}''} = m \}.$$

On the tensor side, define

$$\widehat{T}^{(n,m)}(\mathbb{R}^d) := \text{span} \{ e_{[w]} : |w|_{\mathcal{A}'} = n, \quad |w|_{\mathcal{A}''} = m \},$$

and

$$T_{\mathcal{A}''}((\mathbb{R}^d)) := \prod_{n,m \geq 0} \widehat{T}^{(n,m)}(\mathbb{R}^d).$$

Partial symmetrization is the bidegree-preserving linear map

$$\widehat{\cdot} : T((\mathbb{R}^d)) \longrightarrow T_{\mathcal{A}''}((\mathbb{R}^d)), \quad e_w \longmapsto e_{[w]}.$$

For $\mathbf{a} \in T((\mathbb{R}^d))$, we write $\widehat{\mathbf{a}} := \widehat{\cdot}(\mathbf{a})$. The induced pairing is given by

$$\langle [w], \widehat{\mathbf{a}} \rangle = \sum_{[v]=[w]} \langle v, \mathbf{a} \rangle.$$

The graded transpose of $\widehat{\cdot}$ is the injective map $\widehat{\cdot}^* : \mathbb{R}_{\mathcal{A}''} \langle \mathcal{A} \rangle \rightarrow \mathbb{R} \langle \mathcal{A} \rangle$, characterized by $\widehat{\cdot}^*([w]) = \sum_{[v]=[w]} v$. Consequently, $\langle \alpha, \widehat{\mathbf{a}} \rangle = \langle \widehat{\cdot}^*(\alpha), \mathbf{a} \rangle$. Each distinct ordered representative occurs exactly once in this sum, so no factorial weight is introduced. The map $\widehat{\cdot}^*$ is an adjoint lift, not an inverse of $\widehat{\cdot}$. For $N, M \in \mathbb{N}_0$, define the bigraded truncated partially symmetrized tensor algebra and its dual word space by

$$\widehat{T}^{N,M}(\mathbb{R}^d) := \bigoplus_{n=0}^N \bigoplus_{m=0}^M \widehat{T}^{(n,m)}(\mathbb{R}^d), \quad \mathbb{R}_{\mathcal{A}''}^{N,M} \langle \mathcal{A} \rangle := \bigoplus_{n=0}^N \bigoplus_{m=0}^M \mathbb{R}_{\mathcal{A}''}^{(n,m)} \langle \mathcal{A} \rangle.$$

Equivalently, $\widehat{T}^{N,M}(\mathbb{R}^d) = \widehat{\cdot}(T^{N,M}(\mathbb{R}^d))$. For $\alpha = (\alpha_i)_{i \in \mathcal{A}''} \in \mathbb{N}_0^{\mathcal{A}''}$, we identify α with the common quotient class of all $\mathcal{A}''$ -words containing each letter $\textcolor{brown}{i}$ exactly α_i times, and write

$$|\alpha| := \sum_{\textcolor{brown}{i} \in \mathcal{A}''} \alpha_i.$$

Every quotient word of bidegree (n, m) has a unique *normal form*

$$\alpha_0 \textcolor{blue}{i}_1 \alpha_1 \cdots \textcolor{blue}{i}_n \alpha_n,$$

where $\alpha_0, \dots, \alpha_n \in \mathbb{N}_0^{\mathcal{A}''}$, $\textcolor{blue}{i}_1, \dots, \textcolor{blue}{i}_n \in \mathcal{A}'$, and $|\alpha_0| + \cdots + |\alpha_n| = m$.

1.6. Coordinate changes, quotients, and induced products. In standard coordinates the signature is $\text{Sig}(Y)$, while in shear coordinates it is $\Psi(\text{Sig}(Y))$, since

$$\langle w, \text{Sig}(Y)_{s,t} \rangle_\Psi = \langle w, \Psi(\text{Sig}(Y)_{s,t}) \rangle.$$

The partially symmetrized signature is the quotient tensor series

$$\widehat{\text{Sig}}(Y)_{s,t} := \widehat{\text{Sig}(Y)}_{s,t}.$$

As shown in [2], the tensor-side kernel $\mathcal{I}_{\mathcal{A}''} := \ker(\widehat{\cdot})$ is invariant under Ψ and Ψ^{-1} . Hence, they descend to inverse linear automorphisms $\widehat{\Psi}$ and $\widehat{\Psi}^{-1}$ satisfying

$$\widehat{\Psi}(\widehat{\mathbf{a}}) = \widehat{\Psi(\mathbf{a})}, \quad \widehat{\Psi}^{-1}(\widehat{\mathbf{a}}) = \widehat{\Psi^{-1}(\mathbf{a})}.$$

The corresponding shear pairing on the quotient is therefore defined by

$$\langle \alpha, \widehat{\mathbf{a}} \rangle_\Psi := \langle \alpha, \widehat{\Psi(\mathbf{a})} \rangle, \quad \alpha \in \mathbb{R}_{\mathcal{A}''} \langle \mathcal{A} \rangle.$$

The standard and shear coordinates, before and after partial symmetrization, each carry a tensor product governing the corresponding Chen identity and a dual product governing multiplication of coordinate functionals. The products used below are summarized as follows:

coordinates	group law	dual product
$\text{Sig}(Y)$	$\otimes$	$\sqcup$
$\Psi(\text{Sig}(Y))$	$\odot$	$\bullet$
$\widehat{\text{Sig}}(Y)$	$\widehat{\otimes}$	$\widehat{\sqcup}$
$\widehat{\Psi}(\widehat{\text{Sig}}(Y))$	$\widehat{\odot}$	$\widehat{\bullet}$

The tensor products are related by coordinate transport and passage to the quotient:

$$\mathbf{a} \odot \mathbf{b} := \Psi(\Psi^{-1}(\mathbf{a}) \otimes \Psi^{-1}(\mathbf{b})), \quad \widehat{\mathbf{a}} \widehat{\otimes} \widehat{\mathbf{b}} := \widehat{\mathbf{a} \otimes \mathbf{b}}, \quad \widehat{\mathbf{a}} \widehat{\odot} \widehat{\mathbf{b}} := \widehat{\Psi(\Psi^{-1}(\widehat{\mathbf{a}}) \widehat{\otimes} \Psi^{-1}(\widehat{\mathbf{b}}))}.$$

The products $\sqcup$, $\bullet$, $\widehat{\sqcup}$, and $\widehat{\bullet}$ are the corresponding graded dual products. Thus, for each row of the table, denoting its entries by $\mathbf{S}$, $\otimes$, and $\star$, respectively, we have

$$\mathbf{S}_{s,t} = \mathbf{S}_{s,u} \otimes \mathbf{S}_{u,t}, \quad \langle \alpha, \mathbf{S}_{s,t} \rangle \langle \beta, \mathbf{S}_{s,t} \rangle = \langle \alpha \star \beta, \mathbf{S}_{s,t} \rangle,$$

for $Y \in C^{1\text{-var}}([s, t], \mathbb{R}^d)$, $s \leq u \leq t$, and α, β in the corresponding dual word space. In the second and fourth rows, the standard pairing with the displayed transformed tensor is, equivalently, the Ψ -pairing with the corresponding untransformed signature tensor.

The coordinate changes and partial symmetrization preserve bidegrees, while all tensor and dual products add bidegrees. They therefore descend to the corresponding bigraded truncations. We denote the resulting products by

$$\otimes_{N,M}, \quad \odot_{N,M}, \quad \widehat{\otimes}_{N,M}, \quad \widehat{\odot}_{N,M}, \quad \sqcup_{N,M}, \quad \bullet_{N,M}, \quad \widehat{\sqcup}_{N,M}, \quad \widehat{\bullet}_{N,M}.$$

In the following sections, we revisit these products in their respective coordinate layouts and derive explicit algorithms for their computation.

2. DATA LAYOUTS

For a total-degree truncated tensor, each homogeneous component $(\mathbb{R}^d)^{\otimes r}$ has the canonical dense array layout with shape

$$\underbrace{d \times \cdots \times d}_{r \text{ times}}.$$

For a fixed bidegree (n, m) , however, the $\mathcal{A}'$ - and $\mathcal{A}''$ -letters may occur in different relative positions. One placement pattern, in bidegree $(2, 3)$ for example

$$\bullet \bullet \bullet \bullet \bullet,$$

specifies which two positions are occupied by $\mathcal{A}'$ -letters and which three by $\mathcal{A}''$ -letters. Once the placement is fixed, the corresponding coordinates of the bidegree tensor have the usual dense tensor layout

$$(\mathbb{R}^{|\mathcal{A}'|})^{\otimes n} \otimes (\mathbb{R}^{|\mathcal{A}''|})^{\otimes m}.$$

We therefore store a tensor of bidegree (n, m) as a collection of such dense subtensors, together with an additional axis indexing the possible placements. Under partial symmetrization, the ordered $\mathcal{A}''$ -indices within each block are replaced by their multiplicity multi-indices, and the placement axis is replaced by an axis indexing the resulting multiset placements. The following subsections make these two indexing sets explicit and assign them compact integer ranks.

2.1. Colexicographic ranking. For integers $N \geq 0$ and $0 \leq r \leq N$, let

$$P = \{p_1 < \dots < p_r\} \subseteq \{1, \dots, N\}$$

be an r -element subset. Its colexicographic rank is defined by [3, Theorem L and Eq. (20), p. 360]

$$\text{rk}(P) := \sum_{s=1}^r \binom{p_s - 1}{s}.$$

This defines a bijection from the r -element subsets of $\{1, \dots, N\}$ onto $\{0, \dots, \binom{N}{r} - 1\}$.

We first apply this ranking to the placement patterns of the ordered bidegree blocks. For a fixed bidegree $(n, m) \in \mathbb{N}_0^2$, define the set of $\mathcal{A}'$ -letter placements by

$$\text{Pl}(n, m) := \{P \subseteq \{1, \dots, n + m\} : |P| = n\}.$$

Thus $P \in \text{Pl}(n, m)$ records the positions occupied by the $\mathcal{A}'$ -letters. Applying the colexicographic rank with $L = n + m$ and $r = n$ gives a bijective index on $\text{Pl}(n, m)$, whose cardinality is

$$|\text{Pl}(n, m)| = \binom{n + m}{n}.$$

For the partially symmetrized blocks, the corresponding data are the multiplicities of the $\mathcal{A}''$ -letters in the blocks separated by the $\mathcal{A}'$ -letters. For a fixed bidegree $(n, m) \in \mathbb{N}_0^2$, define the set of multiset placements by

$$\text{MPI}(n, m) := \left\{ (\alpha_0, \dots, \alpha_n) \in (\mathbb{N}_0^{\mathcal{A}''})^{n+1} : \sum_{j=0}^n |\alpha_j| = m \right\}.$$

Thus $\alpha = (\alpha_0, \dots, \alpha_n) \in \text{MPI}(n, m)$ records the multiplicities of the $\mathcal{A}''$ -letters in the $n + 1$ blocks separated by the $\mathcal{A}'$ -letters. Fixing an order on $\mathcal{A}''$, we flatten these multiplicities blockwise as $(a_1, \dots, a_K) \in \mathbb{N}_0^K$, where $K = (n + 1)|\mathcal{A}''|$ and $a_1 + \dots + a_K = m$. The separator positions are then $\{r + a_1 + \dots + a_r : 1 \leq r < K\}$, which gives a bijection between $\text{MPI}(n, m)$ and the $(K - 1)$ -element subsets of $\{1, \dots, m + K - 1\}$. Applying the colexicographic rank, we define

$$(1) \quad \text{rk}_{\text{sym}}(\alpha) := \sum_{r=1}^{K-1} \binom{r - 1 + \sum_{l=1}^r a_l}{r}.$$

Hence rk_{sym} is a bijection from $\text{MPI}(n, m)$ onto $\{0, \dots, |\text{MPI}(n, m)| - 1\}$, where

$$|\text{MPI}(n, m)| = \binom{m + K - 1}{K - 1} = \binom{m + (n + 1)|\mathcal{A}''| - 1}{m}.$$

The rank and its inverse are computed using the standard combination-ranking and unranking algorithms described in [3, Sec. 7.2.1.3, Exercise 17] and [1]. For $K = 1$, set $a_1 = m$. For $K \geq 2$, if unranking yields $\{b_1 < \dots < b_{K-1}\}$, we recover $a_1 = b_1 - 1$, $a_r = b_r - b_{r-1} - 1$ for $2 \leq r < K$, and $a_K = m + K - 1 - b_{K-1}$. We precompute the required rank and inverse-rank tables for every bidegree in the chosen truncation.

2.2. Ordered and multiset placement layouts. Fix an order on each of the alphabets $\mathcal{A}'$ and $\mathcal{A}''$. For $\mathbf{a} \in T^{N,M}(\mathbb{R}^d)$, we store its bidegree- (n, m) component as

$$(2) \quad \mathbf{a}^{(n,m)} \in \mathbb{R}^{|\text{Pl}(n,m)|} \otimes \left(\mathbb{R}^{|\mathcal{A}'|} \right)^{\otimes n} \otimes \left(\mathbb{R}^{|\mathcal{A}''|} \right)^{\otimes m}.$$

The first factor is indexed by $k = \text{rk}(P)$, and we denote the corresponding placement slice by

$$\mathbf{a}_k^{(n,m)} \in \left(\mathbb{R}^{|\mathcal{A}'|} \right)^{\otimes n} \otimes \left(\mathbb{R}^{|\mathcal{A}''|} \right)^{\otimes m}.$$

Its first n dense axes store the $\mathcal{A}'$ -letters from left to right, and its last m dense axes store the $\mathcal{A}''$ -letters from left to right.

Partial symmetrization replaces the ordinary placement axis and the ordered $\mathcal{A}''$ -letter axes by a single multiset-placement axis. For $\hat{\mathbf{a}} \in \hat{T}^{N,M}(\mathbb{R}^d)$, we therefore store its bidegree- (n, m) component as

$$(3) \quad \hat{\mathbf{a}}^{(n,m)} \in \mathbb{R}^{|\text{MPl}(n,m)|} \otimes \left(\mathbb{R}^{|\mathcal{A}'|} \right)^{\otimes n}.$$

For $\boldsymbol{\alpha} = (\alpha_0, \dots, \alpha_n) \in \text{MPl}(n, m)$, the first factor is indexed by $k = \text{rk}_{\text{sym}}(\boldsymbol{\alpha})$, and we denote the corresponding slice by

$$\hat{\mathbf{a}}_k^{(n,m)} \in \left(\mathbb{R}^{|\mathcal{A}'|} \right)^{\otimes n}.$$

Thus the dense axes retain only the $\mathcal{A}'$ -letters, while the multiplicities and block positions of the $\mathcal{A}''$ -letters are encoded by the multiset-placement rank.

3. OPERATIONS IN STANDARD COORDINATES

In this section, we express concatenation, shuffle, and signature computation in the block layouts of [Section 2.2](#). We first treat the ordered coordinates and then their partially symmetrized counterparts. In each case, we separate the combinatorial action on the placement axis from the corresponding operations on the dense axes.

3.1. Ordered coordinates.

3.1.1. Concatenation. For fixed bidegrees (n_1, m_1) and (n_2, m_2) with $n_1 + n_2 \leq N$ and $m_1 + m_2 \leq M$, define

$$C_{n_1, m_1, n_2, m_2} : \mathbb{R}^{|\text{Pl}(n_1, m_1)|} \otimes \mathbb{R}^{|\text{Pl}(n_2, m_2)|} \longrightarrow \mathbb{R}^{|\text{Pl}(n_1 + n_2, m_1 + m_2)|}$$

on basis elements by

$$C_{n_1, m_1, n_2, m_2} (e_{\text{rk}(P_1)} \otimes e_{\text{rk}(P_2)}) := e_{\text{rk}(P_1 \cup (n_1 + m_1 + P_2))},$$

where $n_1 + m_1 + P_2 = \{n_1 + m_1 + p : p \in P_2\}$.

Let $\tau_{n_1, m_1, n_2, m_2}$ be the canonical permutation from

$$\mathbb{R}^{|\text{Pl}(n_1, m_1)|} \otimes \left(\mathbb{R}^{|\mathcal{A}'|} \right)^{\otimes n_1} \otimes \left(\mathbb{R}^{|\mathcal{A}''|} \right)^{\otimes m_1} \otimes \mathbb{R}^{|\text{Pl}(n_2, m_2)|} \otimes \left(\mathbb{R}^{|\mathcal{A}'|} \right)^{\otimes n_2} \otimes \left(\mathbb{R}^{|\mathcal{A}''|} \right)^{\otimes m_2}$$

to

$$\mathbb{R}^{|\text{Pl}(n_1, m_1)|} \otimes \mathbb{R}^{|\text{Pl}(n_2, m_2)|} \otimes \left(\mathbb{R}^{|\mathcal{A}'|} \right)^{\otimes (n_1 + n_2)} \otimes \left(\mathbb{R}^{|\mathcal{A}''|} \right)^{\otimes (m_1 + m_2)}.$$

Lemma 3.1. For $\mathbf{a}, \mathbf{b} \in T^{N,M}(\mathbb{R}^d)$,

$$(\mathbf{a} \otimes_{N,M} \mathbf{b})^{(n,m)} = \sum_{\substack{n_1 + n_2 = n \\ m_1 + m_2 = m}} (C_{n_1, m_1, n_2, m_2} \otimes \text{id}) \tau_{n_1, m_1, n_2, m_2} (\mathbf{a}^{(n_1, m_1)} \otimes \mathbf{b}^{(n_2, m_2)}).$$

Moreover, if $P_2 = \{p_1 < \dots < p_{n_2}\}$, then

$$(4) \quad \text{rk}(P_1 \cup (n_1 + m_1 + P_2)) = \text{rk}(P_1) + \sum_{s=1}^{n_2} \binom{n_1 + m_1 + p_s - 1}{n_1 + s}.$$

Proof. Concatenating words with placements P_1 and P_2 leaves the first placement unchanged and shifts the second by $n_1 + m_1$. The permutation $\tau_{n_1, m_1, n_2, m_2}$ places the two placement axes first and groups the dense $\mathcal{A}'$ - and $\mathcal{A}''$ -axes in the output block order, which gives the first identity.

The elements of P_1 are the first n_1 elements of $P_1 \cup (n_1 + m_1 + P_2)$, while its remaining elements are $n_1 + m_1 + p_s$, $1 \leq s \leq n_2$. The rank formula follows directly from the definition of the colexicographic rank. $\square$

Implementation

For each placement $P_2 = \{p_1 < \dots < p_{n_2}\}$ of rank j_2 , precompute

$$o_{j_2} := \sum_{s=1}^{n_2} \binom{n_1 + m_1 + p_s - 1}{n_1 + s}.$$

The outer loop over the bidegree splittings remains. For one fixed splitting, both placement-rank axes and all dense axes are vectorized. If $\mathbf{V}$ denotes the outer product after applying $\tau_{n_1, m_1, n_2, m_2}$, its two leading axes are indexed by (j_1, j_2) and the block update is

$$1: \mathbf{K} \leftarrow (j_1 + o_{j_2})_{j_1, j_2}$$

$$2: \mathbf{c}^{(n, m)}[\mathbf{K}] += \mathbf{V} \quad \triangleright \text{all target ranks are distinct}$$

Thus there is no runtime loop over j_1 or j_2 . The array $\mathbf{K}$ may be formed by broadcasting the consecutive first-rank axis against the offset array and need not be stored. If the full rank-pair tensor is too large, the second rank axis may be processed in batches; the batch loop is then the only additional loop.

3.1.2. *Shuffle.* Fix input bidegrees (n_1, m_1) and (n_2, m_2) , and write $n = n_1 + n_2$ and $m = m_1 + m_2$. We denote by $\text{Sh}(n_1 + m_1, n_2 + m_2)$ the set of pairs of strictly increasing maps

$$\iota_1: \{1, \dots, n_1 + m_1\} \longrightarrow \{1, \dots, n + m\}, \quad \iota_2: \{1, \dots, n_2 + m_2\} \longrightarrow \{1, \dots, n + m\},$$

with disjoint images whose union is $\{1, \dots, n + m\}$. For two words with $\mathcal{A}'$ placements $P_1 \in \text{Pl}(n_1, m_1)$ and $P_2 \in \text{Pl}(n_2, m_2)$, and shuffle $(\iota_1, \iota_2) \in \text{Sh}(n_1 + m_1, n_2 + m_2)$, the placement of $\mathcal{A}'$ letters in the shuffled word is given by

$$P_{\iota_1, \iota_2}(P_1, P_2) := \iota_1(P_1) \cup \iota_2(P_2) \in \text{Pl}(n, m).$$

Write $\{p_1 < \dots < p_n\} = P_{\iota_1, \iota_2}(P_1, P_2)$. Then the placement of the $\mathcal{A}''$ letters in the shuffled word is given by $\{q_1 < \dots < q_m\} = \{1, \dots, n + m\} \setminus P_{\iota_1, \iota_2}(P_1, P_2)$. Define

$$Q'_{\iota_1, \iota_2}(P_1, P_2) := \{r \in \{1, \dots, n\} : p_r \in \iota_1(P_1)\} \in \text{Pl}(n_1, n_2),$$

and

$$Q''_{\iota_1, \iota_2}(P_1, P_2) := \{s \in \{1, \dots, m\} : q_s \in \iota_1(P_1^c)\} \in \text{Pl}(m_1, m_2),$$

where complements are taken in the corresponding word-position sets. Thus, $Q'_{\iota_1, \iota_2}(P_1, P_2)$ records the positions occupied by the $\mathcal{A}'$ -letters of the first word within the $\mathcal{A}'$ -subsequence of the shuffled word, while $Q''_{\iota_1, \iota_2}(P_1, P_2)$ records the positions occupied by its $\mathcal{A}''$ -letters within the $\mathcal{A}''$ -subsequence.

We then represent the shuffle product by grouping the shuffles according to the induced subsequence placements Q' and Q'' and summing the corresponding output placements. For $Q' \in \text{Pl}(n_1, n_2)$ and $Q'' \in \text{Pl}(m_1, m_2)$, define

$$S_{n_1, m_1, n_2, m_2}^{Q', Q''}: \mathbb{R}^{|\text{Pl}(n_1, m_1)|} \otimes \mathbb{R}^{|\text{Pl}(n_2, m_2)|} \longrightarrow \mathbb{R}^{|\text{Pl}(n, m)|}$$

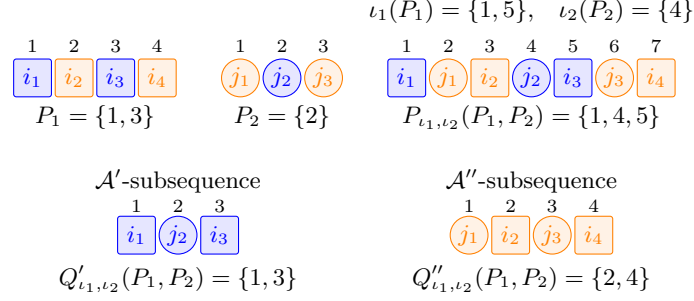


FIGURE 1. The input words have placements $P_1 = \{1, 3\}$ and $P_2 = \{2\}$, and are shuffled by $\iota_1(1, 2, 3, 4) = (1, 3, 5, 7)$ and $\iota_2(1, 2, 3) = (2, 4, 6)$. The resulting shuffled output word has the placement $P_{\iota_1, \iota_2}(P_1, P_2) = \{1, 4, 5\}$ and induced subsequence placements $Q'_{\iota_1, \iota_2}(P_1, P_2) = \{1, 3\}$ and $Q''_{\iota_1, \iota_2}(P_1, P_2) = \{2, 4\}$. Squares and circles distinguish letters from the input words, while blue and orange distinguish $\mathcal{A}'$ - and $\mathcal{A}''$ -letters. The three output placements determine (ι_1, ι_2) uniquely; in the shuffle precomputation, we therefore iterate over compatible triples of output placements.

on basis elements by

$$S_{n_1, m_1, n_2, m_2}^{Q', Q''} (e_{\text{rk}(P_1)} \otimes e_{\text{rk}(P_2)}) := \sum_{\substack{(\iota_1, \iota_2) \in \text{Sh}(n_1 + m_1, n_2 + m_2) \\ Q'_{\iota_1, \iota_2}(P_1, P_2) = Q' \\ Q''_{\iota_1, \iota_2}(P_1, P_2) = Q''}} e_{\text{rk}(P_{\iota_1, \iota_2}(P_1, P_2))}.$$

The remaining part of the shuffle operation is then described by a permutation of the dense axes. Let $\Theta_{Q', Q''}$ be the canonical permutation from

$$\mathbb{R}^{|\text{Pl}(n_1, m_1)|} \otimes (\mathbb{R}^{|\mathcal{A}'|})^{\otimes n_1} \otimes (\mathbb{R}^{|\mathcal{A}''|})^{\otimes m_1} \otimes \mathbb{R}^{|\text{Pl}(n_2, m_2)|} \otimes (\mathbb{R}^{|\mathcal{A}'|})^{\otimes n_2} \otimes (\mathbb{R}^{|\mathcal{A}''|})^{\otimes m_2}$$

to

$$\mathbb{R}^{|\text{Pl}(n_1, m_1)|} \otimes \mathbb{R}^{|\text{Pl}(n_2, m_2)|} \otimes (\mathbb{R}^{|\mathcal{A}'|})^{\otimes n} \otimes (\mathbb{R}^{|\mathcal{A}''|})^{\otimes m}$$

which moves the two placement axes to the front, places the $\mathcal{A}'$ -axes of the first input in the positions Q' , and places its $\mathcal{A}''$ -axes in the positions Q'' . We write $\Theta_{Q', Q''}^{\text{dense}}$ for the corresponding permutation of the dense axes alone, leaving any leading batch axes fixed.

Lemma 3.2. For $\ell_1, \ell_2 \in \mathbb{R}^{N, M} \langle \mathcal{A} \rangle$,

$$(\ell_1 \sqcup_{N, M} \ell_2)^{(n, m)} = \sum_{\substack{n_1 + n_2 = n \\ m_1 + m_2 = m}} \sum_{\substack{Q' \in \text{Pl}(n_1, n_2) \\ Q'' \in \text{Pl}(m_1, m_2)}} \left(S_{n_1, m_1, n_2, m_2}^{Q', Q''} \otimes \text{id} \right) \Theta_{Q', Q''} \left(\ell_1^{(n_1, m_1)} \otimes \ell_2^{(n_2, m_2)} \right).$$

Proof. By bilinearity, it suffices to consider two basis words with placements $P_1 \in \text{Pl}(n_1, m_1)$ and $P_2 \in \text{Pl}(n_2, m_2)$. Each $(\iota_1, \iota_2) \in \text{Sh}(n_1 + m_1, n_2 + m_2)$ contributes one shuffled word. Its placement coordinate is $\text{rk}(P_{\iota_1, \iota_2}(P_1, P_2))$, while its dense $\mathcal{A}'$ - and $\mathcal{A}''$ -axes are obtained by interleaving the dense axes of the two inputs according to $Q'_{\iota_1, \iota_2}(P_1, P_2)$ and $Q''_{\iota_1, \iota_2}(P_1, P_2)$, respectively.

Fix $Q' \in \text{Pl}(n_1, n_2)$ and $Q'' \in \text{Pl}(m_1, m_2)$. The permutation $\Theta_{Q', Q''}$ produces the dense coordinate part of precisely those shuffle terms for which

$$Q'_{\iota_1, \iota_2}(P_1, P_2) = Q', \quad Q''_{\iota_1, \iota_2}(P_1, P_2) = Q''.$$

For the same terms, $S_{n_1, m_1, n_2, m_2}^{Q', Q''}$ sums the corresponding output placement coordinates. Hence

$$\left(S_{n_1, m_1, n_2, m_2}^{Q', Q''} \otimes \text{id} \right) \Theta_{Q', Q''}$$

is exactly the sum of all shuffle terms having the fixed induced placements (Q', Q'') .

Every shuffle pair assigns a single such pair of induced placements, so summing over Q' and Q'' exhausts all shuffle terms without omission or repetition. Finally, summing over all decompositions $n = n_1 + n_2$ and $m = m_1 + m_2$ gives the (n, m) -block of $\ell_1 \sqcup_{N, M} \ell_2$. $\square$

To justify this sparse formula, we first record that the output placement together with the induced placements uniquely determines the input placements and the shuffle.

Lemma 3.3. *The map*

$$\text{Pl}(n_1, m_1) \times \text{Pl}(n_2, m_2) \times \text{Sh}(n_1 + m_1, n_2 + m_2) \longrightarrow \text{Pl}(n, m) \times \text{Pl}(n_1, n_2) \times \text{Pl}(m_1, m_2),$$

given by

$$(P_1, P_2, \iota_1, \iota_2) \longmapsto (P_{\iota_1, \iota_2}(P_1, P_2), Q'_{\iota_1, \iota_2}(P_1, P_2), Q''_{\iota_1, \iota_2}(P_1, P_2)),$$

is a bijection. In particular, for fixed Q' and Q'' , every output placement arises from a unique pair of input placements and a unique shuffle. Consequently, every matrix coefficient of $S_{n_1, m_1, n_2, m_2}^{Q', Q''}$ is either zero or one.

Proof. We construct the inverse map. Let $P = \{p_1 < \dots < p_n\} \in \text{Pl}(n, m)$, write $P^c = \{q_1 < \dots < q_m\}$, and fix $Q' \in \text{Pl}(n_1, n_2)$ and $Q'' \in \text{Pl}(m_1, m_2)$. Define

$$I_1 := \{p_r : r \in Q'\} \cup \{q_s : s \in Q''\}, \quad I_2 := \{1, \dots, n + m\} \setminus I_1.$$

Since $|I_1| = n_1 + m_1$ and $|I_2| = n_2 + m_2$, there are unique strictly increasing maps ι_1 and ι_2 with images I_1 and I_2 . Define

$$P_1 := \iota_1^{-1}(P), \quad P_2 := \iota_2^{-1}(P).$$

The set P_1 consists of the n_1 indices corresponding to $\{p_r : r \in Q'\}$, while P_2 consists of the n_2 indices corresponding to the remaining positions in P . Hence $P_1 \in \text{Pl}(n_1, m_1)$ and $P_2 \in \text{Pl}(n_2, m_2)$. By construction,

$$P_{\iota_1, \iota_2}(P_1, P_2) = P, \quad Q'_{\iota_1, \iota_2}(P_1, P_2) = Q', \quad Q''_{\iota_1, \iota_2}(P_1, P_2) = Q''.$$

$\square$

For $0 \leq r' < |\text{Pl}(n_1, n_2)|$ and $0 \leq r'' < |\text{Pl}(m_1, m_2)|$, let

$$S_{n_1, m_1, n_2, m_2}^{r', r''} \subseteq \{0, \dots, |\text{Pl}(n_1, m_1)| - 1\} \times \{0, \dots, |\text{Pl}(n_2, m_2)| - 1\} \times \{0, \dots, |\text{Pl}(n, m)| - 1\}$$

be the set of all triples (j_1, j_2, k) such that the input placements of ranks j_1 and j_2 produce the output placement of rank k with induced placements $Q' = \text{rk}^{-1}(r')$ and $Q'' = \text{rk}^{-1}(r'')$. Equivalently,

$$S_{n_1, m_1, n_2, m_2}^{Q', Q''}(e_{j_1} \otimes e_{j_2}) = \sum_{\substack{0 \leq k < |\text{Pl}(n, m)| \\ (j_1, j_2, k) \in S_{n_1, m_1, n_2, m_2}^{r', r''}}} e_k.$$

By [Lemma 3.3](#), no multiplicities are required. These sets are enumerated by [Algorithm 1](#).

Implementation

The nested placement loops in [Algorithm 1](#) are precomputation only. At runtime, the loops are over the bidegree splittings and the keys (r', r'') ; there is no loop over output placement ranks. For one fixed key, write J_1 and J_2 for the two stored input-rank columns. The complete update is

$$1: \mathbf{V} \leftarrow \text{batchouter} \left(\ell_1^{(n_1, m_1)}[J_1], \ell_2^{(n_2, m_2)}[J_2] \right)$$

Algorithm 1 Placement-rank lists for the shuffle**Require:** bidegrees (n_1, m_1) and (n_2, m_2) **Ensure:** the lists $S_{n_1, m_1, n_2, m_2}^{r', r''}$

```

1:  $n \leftarrow n_1 + n_2, m \leftarrow m_1 + m_2$ 
2: for all  $Q' \in \text{Pl}(n_1, n_2)$  with increasing rank  $r'$  do
3:   for all  $Q'' \in \text{Pl}(m_1, m_2)$  with increasing rank  $r''$  do
4:      $S_{n_1, m_1, n_2, m_2}^{r', r''} \leftarrow []$ 
5:   end for
6: end for
7: for all  $P = \{p_1 < \dots < p_n\} \in \text{Pl}(n, m)$  with increasing rank  $k$  do
8:   write  $P^c = \{q_1 < \dots < q_m\}$ 
9:   for all  $Q' \in \text{Pl}(n_1, n_2)$  with increasing rank  $r'$  do
10:    for all  $Q'' \in \text{Pl}(m_1, m_2)$  with increasing rank  $r''$  do
11:       $I_1 \leftarrow \{p_r : r \in Q'\} \cup \{q_s : s \in Q''\}$ 
12:       $I_2 \leftarrow \{1, \dots, n + m\} \setminus I_1$ 
13:      write  $I_1 = \{h_1 < \dots < h_{n_1+m_1}\}$  and  $I_2 = \{\bar{h}_1 < \dots < \bar{h}_{n_2+m_2}\}$ 
14:       $P_1 \leftarrow \{a : h_a \in P\}$ 
15:       $P_2 \leftarrow \{b : \bar{h}_b \in P\}$ 
16:      append  $(\text{rk}(P_1), \text{rk}(P_2), k)$  to  $S_{n_1, m_1, n_2, m_2}^{r', r''}$ 
17:    end for
18:  end for
19: end for

```

```

2:  $\mathbf{V} \leftarrow \Theta_{Q', Q''}^{\text{dense}} \mathbf{V}$ 
3:  $\mathbf{c}^{(n, m)} += \mathbf{V}$ 

```

Here batchouter forms one outer product for every row of the gathered rank axis. The rows are already in increasing output-rank order, so the final addition is contiguous and no output-rank or coefficient array is stored. If necessary, the rows may be processed in contiguous batches without changing the computation.

3.1.3. *Computing signatures.* Let $z = (z', z'') \in \mathbb{R}^{|\mathcal{A}'|} \oplus \mathbb{R}^{|\mathcal{A}''|}$. The bidegree-truncated product

$$\mathbf{b}_0 = \pi^{N, M}(\mathbf{a} \otimes \exp_{\otimes}(z))$$

is computed by the Horner recursion in [Algorithm 2](#), directly on the stored bidegree blocks. Set $\kappa_{n,0} := 0$ and, for $m > 0$, $\kappa_{n,m} := |\text{Pl}(n, m - 1)|$.

The outer loop over q and the active bidegrees in [Algorithm 2](#) is common to all four signature computations. With $\hat{\otimes}$, $\odot$, or $\hat{\odot}$, only the fixed-bidegree formula for right multiplication by z changes. In each case, repeated substitution in the corresponding unpruned recursion gives the appropriate tensor exponential through degree $N + M$. Since right multiplication by z raises the total degree by one, discarding inactive blocks cannot change any block retained by $\pi^{N, M}$. We use this product–Horner argument below without repeating it.

Implementation

The runtime loops over the Horner depth q and the active bidegrees remain, but the placement-rank loops in [Algorithm 2](#) are vectorized. For a fixed (q, n, m) , the complete block update is

```

1:  $\mathbf{b}_{q-1}^{(n, m)} \leftarrow \mathbf{a}^{(n, m)}$ 
2: if  $m > 0$  then

```

Algorithm 2 Horner recursion in ordered standard coordinates**Require:** $\mathbf{a} \in T^{N,M}(\mathbb{R}^d)$ and $z = (z', z'') \in \mathbb{R}^d$ **Ensure:** $\mathbf{b}_0 = \pi^{N,M}(\mathbf{a} \otimes \exp_{\otimes}(z))$

```

1:  $\mathbf{b}_{N+M}^{(0,0)} \leftarrow \mathbf{a}^{(0,0)}$ 
2: All other blocks of  $\mathbf{b}_{N+M}$  are absent
3: for  $q = N + M, \dots, 1$  do
4:    $\mathbf{b}_{q-1}^{(0,0)} \leftarrow \mathbf{a}^{(0,0)}$ 
5:   for all  $(n, m)$  such that  $0 \leq n \leq N$ ,  $0 \leq m \leq M$ , and  $1 \leq n + m \leq N + M - q + 1$  do
      $// (*)$ 
6:      $\mathbf{b}_{q-1}^{(n,m)} \leftarrow \mathbf{a}^{(n,m)}$ 
7:     if  $m > 0$  then
8:       for  $k = 0, \dots, \kappa_{n,m} - 1$  do
9:          $(\mathbf{b}_{q-1}^{(n,m)})_k \leftarrow (\mathbf{b}_{q-1}^{(n,m)})_k + \frac{1}{q} (\mathbf{b}_q^{(n,m-1)})_k \otimes z''$ 
10:       end for
11:     end if
12:     if  $n > 0$  then
13:       for  $k = 0, \dots, |\text{Pl}(n-1, m)| - 1$  do
14:          $(\mathbf{b}_{q-1}^{(n,m)})_{k+\kappa_{n,m}} \leftarrow (\mathbf{b}_{q-1}^{(n,m)})_{k+\kappa_{n,m}} + \frac{1}{q} \tau_{n-1,m,1,0} \left( (\mathbf{b}_q^{(n-1,m)})_k \otimes z' \right)$ 
15:       end for
16:     end if
17:   end for
18: end for
19: return  $\mathbf{b}_0$ 

```

(*) For fixed q , the active bidegrees may be traversed in any order and their computation may be parallelized.

```

3:    $\mathbf{b}_{q-1}^{(n,m)}[0 : \kappa_{n,m}] += q^{-1} \mathbf{b}_q^{(n,m-1)} \otimes z''$ 
4: end if
5: if  $n > 0$  then
6:    $\mathbf{b}_{q-1}^{(n,m)}[\kappa_{n,m} : |\text{Pl}(n, m)|] += q^{-1} \tau_{n-1,m,1,0} \left( \mathbf{b}_q^{(n-1,m)} \otimes z' \right)$ 
7: end if

```

Both assignments act on contiguous rank ranges and are vectorized over all rank and dense axes. The Horner update itself does not consult any placement map, and only $\mathbf{b}_q$ and $\mathbf{b}_{q-1}$ are live numerical states.

If Y is piecewise linear on $s = t_0 < \dots < t_J = t$, with increments $\Delta Y_j = Y_{t_{j-1}, t_j}$, we set $\mathbf{S}_0 = e_\emptyset$ and apply the recursion successively for

$$\mathbf{S}_j = \pi^{N,M}(\mathbf{S}_{j-1} \otimes \exp_{\otimes}(\Delta Y_j)), \quad j = 1, \dots, J.$$

Chen's identity then gives $\mathbf{S}_J = \pi^{N,M} \text{Sig}(Y)_{s,t}$. The same partitionwise assembly applies in the remaining three settings, using the corresponding product and fixed-bidegree update described below.

3.2. Partially symmetrized coordinates.

3.2.1. Concatenation. For $\alpha = (\alpha_0, \dots, \alpha_{n_1}) \in \text{MPI}(n_1, m_1)$ and $\beta = (\beta_0, \dots, \beta_{n_2}) \in \text{MPI}(n_2, m_2)$, define

$$\alpha.\beta := (\alpha_0, \dots, \alpha_{n_1-1}, \alpha_{n_1} + \beta_0, \beta_1, \dots, \beta_{n_2}) \in \text{MPI}(n_1 + n_2, m_1 + m_2).$$

This merges the two adjacent $\mathcal{A}''$ -blocks at the concatenation boundary. For $n = n_1 + n_2$ and $m = m_1 + m_2$, define the linear map

$$\widehat{C}_{n_1, m_1, n_2, m_2} : \mathbb{R}^{|\text{MPI}(n_1, m_1)|} \otimes \mathbb{R}^{|\text{MPI}(n_2, m_2)|} \longrightarrow \mathbb{R}^{|\text{MPI}(n, m)|}$$

on basis elements by

$$\widehat{C}_{n_1, m_1, n_2, m_2} (e_{\text{rk}_{\text{sym}}(\alpha)} \otimes e_{\text{rk}_{\text{sym}}(\beta)}) := e_{\text{rk}_{\text{sym}}(\alpha.\beta)}.$$

Let $\widehat{\tau}_{n_1, m_1, n_2, m_2}$ be the canonical permutation of tensor factors which sends

$$\mathbb{R}^{|\text{MPI}(n_1, m_1)|} \otimes \left(\mathbb{R}^{|\mathcal{A}'|} \right)^{\otimes n_1} \otimes \mathbb{R}^{|\text{MPI}(n_2, m_2)|} \otimes \left(\mathbb{R}^{|\mathcal{A}'|} \right)^{\otimes n_2}$$

to

$$\mathbb{R}^{|\text{MPI}(n_1, m_1)|} \otimes \mathbb{R}^{|\text{MPI}(n_2, m_2)|} \otimes \left(\mathbb{R}^{|\mathcal{A}'|} \right)^{\otimes n}.$$

Then, in complete analogy with [Lemma 3.1](#), we obtain the following block formula.

Lemma 3.4. For $\widehat{\mathbf{a}}, \widehat{\mathbf{b}} \in \widehat{T}^{N, M}(\mathbb{R}^d)$,

$$(\widehat{\mathbf{a}} \otimes_{N, M} \widehat{\mathbf{b}})^{(n, m)} = \sum_{\substack{n_1 + n_2 = n \\ m_1 + m_2 = m}} \left(\widehat{C}_{n_1, m_1, n_2, m_2} \otimes \text{id} \right) \widehat{\tau}_{n_1, m_1, n_2, m_2} \left(\widehat{\mathbf{a}}^{(n_1, m_1)} \otimes \widehat{\mathbf{b}}^{(n_2, m_2)} \right).$$

In contrast to the ordered case, the map from pairs of input ranks to the target rank is generally not injective, since the merged boundary block $\alpha_{n_1} + \beta_0$ does not retain how its entries were split between the two inputs. Thus, $\widehat{C}_{n_1, m_1, n_2, m_2}$ is a genuine sparse accumulation map rather than an injective re-indexing. Moreover, unlike [\(4\)](#), the target rank does not decompose as the first input rank plus an offset depending only on the second input. The interaction of the two inputs is, however, confined to the boundary blocks, and the rank arithmetic remains fully explicit. Indeed, recall from the definition of rk_{sym} in [\(1\)](#) that, using the fixed order on $\mathcal{A}''$, each $\alpha \in \text{MPI}(n_1, m_1)$ is flattened blockwise as $(a_1, \dots, a_{(n_1+1)|\mathcal{A}''|})$. We define

$$\text{rk}^{\text{pre}}(\alpha) := \sum_{r=1}^{n_1|\mathcal{A}''|} \binom{r-1 + \sum_{\ell=1}^r a_\ell}{r}.$$

Thus, rk^{pre} collects the first $n_1|\mathcal{A}''|$ terms of rk_{sym} . Write $\mathcal{A}'' = \{i_1, \dots, i_{|\mathcal{A}''|}\}$ in its fixed order. For $\mu, \nu \in \mathbb{N}_0^{\mathcal{A}''}$ and $q \in \mathbb{N}_0$, define

$$\text{rk}_{n_1}^{\text{bd}}(\mu, \nu, q) := \sum_{u=1}^{|\mathcal{A}''|} \binom{n_1|\mathcal{A}''| + u - 1 + q + \sum_{v=1}^u (\mu_{i_v} + \nu_{i_v})}{n_1|\mathcal{A}''| + u}.$$

Likewise, if $(b_1, \dots, b_{(n_2+1)|\mathcal{A}''|})$ is the blockwise flattening of $\beta \in \text{MPI}(n_2, m_2)$, define

$$\text{rk}_{n_1, m_1}^{\text{suf}}(\beta) := \sum_{u=|\mathcal{A}''|+1}^{(n_2+1)|\mathcal{A}''|} \binom{n_1|\mathcal{A}''| + u - 1 + m_1 + \sum_{\ell=1}^u b_\ell}{n_1|\mathcal{A}''| + u}.$$

Lemma 3.5. For $n = n_1 + n_2$, $m = m_1 + m_2$, $\alpha \in \text{MPI}(n_1, m_1)$, and $\beta \in \text{MPI}(n_2, m_2)$, we have

$$\text{rk}_{\text{sym}}(\alpha.\beta) = \text{rk}^{\text{pre}}(\alpha) + \text{rk}_{n_1}^{\text{bd}}(\alpha_{n_1}, \beta_0, m_1 - |\alpha_{n_1}|) + \text{rk}_{n_1, m_1}^{\text{suf}}(\beta) - \binom{(n+1)|\mathcal{A}''| - 1 + m}{(n+1)|\mathcal{A}''|}.$$

Proof. Let $(g_1, \dots, g_{(n+1)|\mathcal{A}''|})$ be the flattening of $\alpha.\beta$ and set $S_r := \sum_{\ell=1}^r g_\ell$. For $r \leq n_1|\mathcal{A}''|$, the flattening agrees with that of α , so $S_r = \sum_{\ell=1}^r a_\ell$. For $r = n_1|\mathcal{A}''| + u$ with $1 \leq u \leq |\mathcal{A}''|$, the partial sum consists of the weight of the prefix of α and the first u entries of the merged boundary block, so

$$S_r = m_1 - |\alpha_{n_1}| + \sum_{v=1}^u (\alpha_{n_1, i_v} + \beta_{0, i_v}).$$

For $r = n_1|\mathcal{A}''| + u$ with $|\mathcal{A}''| < u \leq (n_2 + 1)|\mathcal{A}''|$, the partial sum contains the full weight m_1 of α and the first u entries of the flattening of β , so $S_r = m_1 + \sum_{\ell=1}^u b_\ell$. Summing $\binom{r-1+S_r}{r}$ over the three regions yields the three terms on the right-hand side. Together, these regions cover $r = 1, \dots, (n+1)|\mathcal{A}''|$, whereas rk_{sym} omits the final determined index $r = (n+1)|\mathcal{A}''|$. Since $S_{(n+1)|\mathcal{A}''|} = m$, its contribution gives the final subtraction. $\square$

Implementation

The outer loop over the bidegree splittings remains. For one fixed splitting, all input-rank pairs and all dense axes are vectorized. Store only the target-rank array

$$K(j_1, j_2) := \text{rk}_{\text{sym}}(\alpha.\beta)$$

in lexicographic input-pair order. The runtime update is

- 1: $\mathbf{V} \leftarrow \hat{\tau}_{n_1, m_1, n_2, m_2}(\hat{\mathbf{a}}^{(n_1, m_1)} \otimes \hat{\mathbf{b}}^{(n_2, m_2)})$
- 2: flatten the two rank axes of $\mathbf{V}$
- 3: $\hat{\mathbf{c}}^{(n, m)} \leftarrow \text{scatteradd}(\hat{\mathbf{c}}^{(n, m)}, K, \mathbf{V})$

Here $\text{scatteradd}(\mathbf{C}, K, \mathbf{V})$ adds row j of $\mathbf{V}$ to row K_j of $\mathbf{C}$, so repeated target ranks are accumulated directly. Thus there is no runtime loop over (j_1, j_2) and no stored ordering, segment offsets, or list of distinct targets. If the pair axis is too large to materialize, it may be processed in batches with partial updates scattered to the same target ranks.

3.2.2. Shuffle. Although the quotient shuffle product $\hat{\sqcup}_{N, M}$ can be computed recursively in the partially symmetrized block layout, we do not compile this recursion directly in partially symmetrized standard coordinates. Already for $i \in \mathcal{A}''$ and $k, \ell \in \mathcal{A}'$, we have

$$([i]k) \hat{\sqcup} ([i]\ell) = [i]k[i]\ell + 2[ii]k\ell + 2[ii]\ell k + [i]\ell[i]k.$$

The coefficients arise because different shuffle placements produce the same partially symmetrized basis element. In higher bidegrees, the $\mathcal{A}''$ -blocks may additionally split across several gaps created by an interleaving of the $\mathcal{A}'$ -skeletons, and all decompositions producing the same output basis element must be accumulated.

By contrast, the corresponding product in partially symmetrized shear coordinates is simply

$$([i]k) \hat{\bullet} ([i]\ell) = [i]k[i]\ell + [i]\ell[i]k.$$

Indeed, $\hat{\bullet}$ only interleaves the two units $[i]k$ and $[i]\ell$; no $\mathcal{A}''$ -block is split. More generally, only the $\mathcal{A}'$ -skeletons are interleaved, while the two terminal $\mathcal{A}''$ -blocks are merged with a single multinomial coefficient, as shown in [Lemma 4.5](#).

This suggests carrying out the entire computation on the dual side in partially symmetrized shear coordinates. Under the coordinate transformation, the dual coordinate map $(\hat{\Psi}^{-1})^*$ is characterized by $\langle (\hat{\Psi}^{-1})^*(\ell), \hat{\mathbf{a}} \rangle := \langle \ell, \hat{\Psi}^{-1}(\hat{\mathbf{a}}) \rangle$. Shuffle products then satisfy

$$(\hat{\Psi}^{-1})^*(\ell_1 \hat{\sqcup}_{N, M} \ell_2) = (\hat{\Psi}^{-1})^*(\ell_1) \hat{\bullet}_{N, M} (\hat{\Psi}^{-1})^*(\ell_2),$$

while right concatenation by $i \in \mathcal{A}'$ is preserved¹:

$$(\widehat{\Psi}^{-1})^*(\ell i) = (\widehat{\Psi}^{-1})^*(\ell) i.$$

3.2.3. Computing signatures. Let $z = (z', z'') \in \mathbb{R}^{|\mathcal{A}'|} \oplus \mathbb{R}^{|\mathcal{A}''|}$. In direct analogy with the ordered-coordinate recursion in [Algorithm 2](#), write $\exp_{\widehat{\otimes}}(z) := \widehat{\exp_{\otimes}}(z)$. The bidegree-truncated product

$$\widehat{\mathbf{b}}_0 = \widehat{\mathbf{a}} \widehat{\otimes}_{N,M} \exp_{\widehat{\otimes}}(z)$$

is computed by the outer recursion of [Algorithm 2](#), with $\widehat{\otimes}$ in place of $\otimes$ and the partially symmetrized fixed-bidegree update below.

Implementation

The runtime loops over q and the active bidegrees remain. For the z'' -contribution, vectorize jointly over $i \in \mathcal{A}''$, the complete input-rank axis, and all dense $\mathcal{A}'$ -axes. Let $\delta_i \in \mathbb{N}_0^{\mathcal{A}''}$ be the unit multi-index at i . If $(\alpha_0, \dots, \alpha_n) = \text{rk}_{\text{sym}}^{-1}(k) \in \text{MPI}(n, m-1)$, precompute

$$K''_{n,m}(i, k) := \text{rk}_{\text{sym}}(\alpha_0, \dots, \alpha_n + \delta_i).$$

Flatten the pairs (i, k) in lexicographic order. The direct plan stores the resulting target array $K''_{n,m}$. For the destination-ordered plan, store the increasing array H of distinct target ranks, one primary edge index in E_{pri} for each entry of H , and the remaining edge indices and targets in E_{col} and K_{col} . The fixed-bidegree update is

```

1:  $\widehat{\mathbf{b}}_{q-1}^{(n,m)} \leftarrow \widehat{\mathbf{a}}^{(n,m)}$ 
2: if  $m > 0$  then
3:   form the flattened edge array  $\mathbf{V}_{(i,k)} \leftarrow q^{-1} z''_i (\widehat{\mathbf{b}}_q^{(n,m-1)})_k$ 
4:   if the direct plan is retained then
5:      $\widehat{\mathbf{b}}_{q-1}^{(n,m)} \leftarrow \text{scatteradd}(\widehat{\mathbf{b}}_{q-1}^{(n,m)}, K''_{n,m}, \mathbf{V})$ 
6:   else
7:      $\widehat{\mathbf{b}}_{q-1}^{(n,m)}[H] \mathrel{+}= \mathbf{V}[E_{\text{pri}}]$ 
8:      $\widehat{\mathbf{b}}_{q-1}^{(n,m)} \leftarrow \text{scatteradd}(\widehat{\mathbf{b}}_{q-1}^{(n,m)}, K_{\text{col}}, \mathbf{V}[E_{\text{col}}])$ 
9:   end if
10: end if
11: if  $n > 0$  then
12:    $\widehat{\mathbf{b}}_{q-1}^{(n,m)}[|\text{MPI}(n, m)| - |\text{MPI}(n-1, m)| : |\text{MPI}(n, m)|] \mathrel{+}= q^{-1} \widehat{\mathbf{b}}_q^{(n-1,m)} \otimes z'$ 
13: end if

```

There is no runtime loop over letters or multiset-placement ranks. The direct plan retains only $K''_{n,m}$; the destination-ordered plan retains H , E_{pri} , E_{col} , and K_{col} . Only two Horner states are live.

4. OPERATIONS IN SHEAR COORDINATES

In this section, we express the coordinate transformations, transported concatenation products, signature computations, and dual products. Since Ψ and Ψ^{-1} are graded, they commute with π_K and restrict to $T^K(\mathbb{R}^d)$. Hence, under total-degree truncation, computations in ordered shear coordinates are performed directly on the canonical dense homogeneous components and require neither bidegree blocks nor placement ranks. More precisely, enumerate the words of length r as $w_0, \dots, w_{d^r-1}$ in the lexicographic order induced by the fixed alphabet order and, for $L \in \{\Psi, \Psi^{-1}\}$, precompute

$$D_r^L(k, \ell) := \langle w_k, L(e_{w_\ell}) \rangle.$$

¹Right concatenation by $i \in \mathcal{A}''$, corresponding to integration against X^i , must instead be transported to shear coordinates by integration by parts $(\widehat{\Psi}^{-1})^* \circ \mathcal{R}_i \circ \widehat{\Psi}^*$, where $\mathcal{R}_i(\ell) = \ell i$.

If $x_\ell = \langle w_\ell, \mathbf{a}^{(r)} \rangle$, then the flattened coefficient vector of $L(\mathbf{a}^{(r)})$ is $D_r^L x$ and is reshaped back to the canonical r -axis array. The nonzero row, column, and coefficient lists of D_r^L may be retained instead of the dense matrix when this uses less memory. A finite maximum degree is required because these degree-wise maps are precomputed. The remainder of the section gives the corresponding formulas in the bidegree block layouts of [Section 2.2](#). As in [Section 3](#), we first treat the ordered coordinates and then their partially symmetrized counterparts.

4.1. Ordered coordinates.

4.1.1. *Concatenation.* Since Ψ and Ψ^{-1} preserve bidegrees, the transported product from [Section 1.6](#) restricts to

$$\mathbf{a} \odot_{N,M} \mathbf{b} = \Psi \left(\Psi^{-1}(\mathbf{a}) \otimes_{N,M} \Psi^{-1}(\mathbf{b}) \right), \quad \mathbf{a}, \mathbf{b} \in T^{N,M}(\mathbb{R}^d).$$

4.1.2. *Shuffle.* Since $\bullet$ adds bidegrees, it induces a truncated product

$$\bullet_{N,M} : \mathbb{R}^{N,M} \langle \mathcal{A} \rangle \times \mathbb{R}^{N,M} \langle \mathcal{A} \rangle \longrightarrow \mathbb{R}^{N,M} \langle \mathcal{A} \rangle.$$

Recall from [\[2, Proposition 3.7\]](#) that, for $w, v \in \mathcal{W}_\mathcal{A}$, $r, q \in \mathcal{W}_{\mathcal{A}''}$, and $i, j \in \mathcal{A}'$, the product is recursively determined by

$$(5) \quad \begin{cases} r \bullet q = r \sqcup q, \\ (w i r) \bullet q = w i (r \sqcup q), \\ r \bullet (v j q) = v j (r \sqcup q), \\ (w i r) \bullet (v j q) = ((w i) \bullet v) j (r \sqcup q) + (w \bullet (v j)) i (r \sqcup q). \end{cases}$$

We now express the product in the ordered block layout. Fix bidegrees (n_1, m_1) and (n_2, m_2) , and write $n = n_1 + n_2$ and $m = m_1 + m_2$. We identify two input placements $P_1 \in \text{Pl}(n_1, m_1)$ and $P_2 \in \text{Pl}(n_2, m_2)$ by their corresponding $\mathcal{A}''$ -block lengths $\delta p_{l,1}, \dots, \delta p_{l,n_l+1}$, characterized by

$$P_l \cup \{n_l + m_l + 1\} = \left\{ t + \sum_{s=1}^t \delta p_{l,s} : t = 1, \dots, n_l + 1 \right\}, \quad l \in \{1, 2\}.$$

To interleave the two ordered sequences of $\mathcal{A}''$ -block/ $\mathcal{A}'$ -letter units, let

$$Q' = \{q_1 < \dots < q_{n_1}\} \in \text{Pl}(n_1, n_2),$$

and write

$$\{\bar{q}_1 < \dots < \bar{q}_{n_2}\} = \{1, \dots, n\} \setminus Q'.$$

The placement Q' specifies the order of the two sequences of $\mathcal{A}'$ -letters. Define the merged block lengths $\delta p_1, \dots, \delta p_n$ by

$$\delta p_{q_s} = \delta p_{1,s}, \quad 1 \leq s \leq n_1, \quad \delta p_{\bar{q}_t} = \delta p_{2,t}, \quad 1 \leq t \leq n_2.$$

The resulting output placement is

$$P_{Q'}(P_1, P_2) = \left\{ t + \sum_{u=1}^t \delta p_u : t = 1, \dots, n \right\} \in \text{Pl}(n, m).$$

It remains to shuffle the two terminal $\mathcal{A}''$ -blocks. For

$$R \in \text{Pl}(\delta p_{1,n_1+1}, \delta p_{2,n_2+1}),$$

define

$$\begin{aligned} Q''_{Q',R}(P_1, P_2) = & \bigcup_{s=1}^{n_1} \left(\sum_{u=1}^{q_s-1} \delta p_u + \{1, \dots, \delta p_{1,s}\} \right) \\ & \cup \left(\sum_{u=1}^n \delta p_u + R \right) \in \text{Pl}(m_1, m_2), \end{aligned}$$

where $c + A = \{c + a : a \in A\}$ and $\{1, \dots, 0\} = \emptyset$. Thus, $Q''_{Q',R}(P_1, P_2)$ records the positions occupied by the $\mathcal{A}''$ -letters of the first input within the $\mathcal{A}'$ -subsequence of the output.

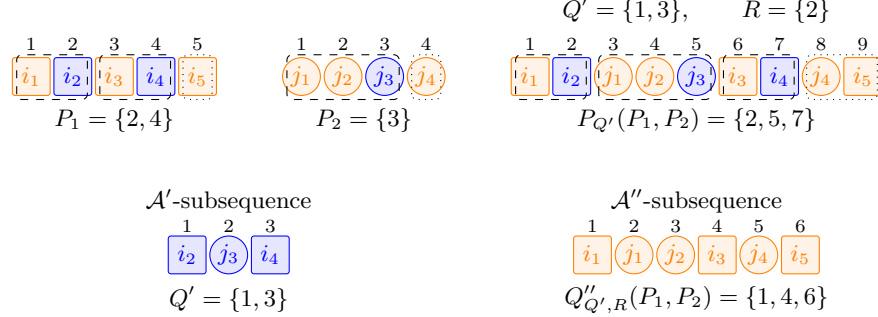


FIGURE 2. The input words have placements $P_1 = \{2, 4\}$ and $P_2 = \{3\}$. The placement $Q' = \{1, 3\}$ orders the two $\mathcal{A}''$ -block/ $\mathcal{A}'$ -letter units of the first input around the unit of the second input, while $R = \{2\}$ shuffles the two terminal $\mathcal{A}''$ -blocks. The resulting output word has placement $P_{Q'}(P_1, P_2) = \{2, 5, 7\}$, and the induced $\mathcal{A}''$ -subsequence placement is $Q''_{Q',R}(P_1, P_2) = \{1, 4, 6\}$. Squares and circles distinguish letters from the two input words, while blue and orange distinguish $\mathcal{A}'$ - and $\mathcal{A}''$ -letters.

We represent $\bullet_{N,M}$ by grouping its terms according to the induced subsequence placements Q' and Q'' and summing the corresponding output placements. For $Q' \in \text{Pl}(n_1, n_2)$ and $Q'' \in \text{Pl}(m_1, m_2)$, define

$$S_{n_1, m_1, n_2, m_2}^{\bullet, Q', Q''} : \mathbb{R}^{|\text{Pl}(n_1, m_1)|} \otimes \mathbb{R}^{|\text{Pl}(n_2, m_2)|} \longrightarrow \mathbb{R}^{|\text{Pl}(n, m)|}$$

on basis elements by

$$S_{n_1, m_1, n_2, m_2}^{\bullet, Q', Q''} (e_{\text{rk}(P_1)} \otimes e_{\text{rk}(P_2)}) = \sum_{\substack{R \in \text{Pl}(\delta p_{1, n_1+1}, \delta p_{2, n_2+1}) \\ Q''_{Q', R}(P_1, P_2) = Q''}} e_{\text{rk}(P_{Q'}(P_1, P_2))}.$$

The dense axes are permuted by the same canonical maps $\Theta_{Q', Q''}$ as in [Section 3.1.2](#) for the ordered standard-coordinate shuffle.

Lemma 4.1. For $\ell_1, \ell_2 \in \mathbb{R}^{N, M} \langle \mathcal{A} \rangle$,

$$(\ell_1 \bullet_{N, M} \ell_2)^{(n, m)} = \sum_{\substack{n_1 + n_2 = n \\ m_1 + m_2 = m}} \sum_{\substack{Q' \in \text{Pl}(n_1, n_2) \\ Q'' \in \text{Pl}(m_1, m_2)}} \left(S_{n_1, m_1, n_2, m_2}^{\bullet, Q', Q''} \otimes \text{id} \right) \Theta_{Q', Q''} \left(\ell_1^{(n_1, m_1)} \otimes \ell_2^{(n_2, m_2)} \right).$$

For each bidegree splitting, the inner sum is supported on

$$\left\{ (Q', Q''_{Q', R}(P_1, P_2)) : P_l \in \text{Pl}(n_l, m_l), \quad l \in \{1, 2\}, \quad Q' \in \text{Pl}(n_1, n_2), \quad R \in \text{Pl}(\delta p_{1, n_1+1}, \delta p_{2, n_2+1}) \right\}.$$

Proof. We use the notation introduced before the lemma. By bilinearity, it suffices to consider two basis words with placements P_1 and P_2 . In the recursion (5), each $\mathcal{A}''$ -block preceding an $\mathcal{A}'$ -letter remains attached to that letter. The two ordered sequences of such units are therefore interleaved according to a unique $Q' \in \text{Pl}(n_1, n_2)$. The merged block lengths are $\delta p_1, \dots, \delta p_n$, so the t -th $\mathcal{A}'$ -letter occurs at position $t + \sum_{u=1}^t \delta p_u$. This gives the output placement $P_{Q'}(P_1, P_2)$.

The recursion applies the ordinary shuffle only to the two terminal $\mathcal{A}''$ -blocks. Its terms are indexed by

$$R \in \text{Pl}(\delta p_{1, n_1+1}, \delta p_{2, n_2+1}),$$

and the positions occupied by all $\mathcal{A}'$ -letters of the first input are precisely $Q''_{Q',R}(P_1, P_2)$. Hence, the dense axes of the corresponding term are permuted by

$$\Theta_{Q', Q''_{Q',R}(P_1, P_2)}.$$

Grouping the terms according to the resulting pair (Q', Q'') gives the displayed formula and its stated support. The recursion partitions the unit interleavings according to the final $\mathcal{A}'$ -letter, and the ordinary terminal shuffle contains each R once. Thus, for fixed input placements P_1 and P_2 , every pair (Q', R) contributes exactly once, with coefficient one. $\square$

As in the standard-coordinate shuffle, the output placement and the two induced subsequence placements determine the input data uniquely.

Lemma 4.2. *For fixed bidegrees, the map*

$$(P_1, P_2, Q', R) \mapsto (P_{Q'}(P_1, P_2), Q', Q''_{Q',R}(P_1, P_2))$$

is injective, where $P_l \in \text{Pl}(n_l, m_l)$, $Q' \in \text{Pl}(n_1, n_2)$, and

$$R \in \text{Pl}(\delta p_{1, n_1+1}, \delta p_{2, n_2+1}).$$

Consequently, every matrix coefficient of $S_{n_1, m_1, n_2, m_2}^{\bullet, Q', Q''}$ is either zero or one.

Proof. We reverse the construction above step by step. From $P_{Q'}(P_1, P_2) = \{p_1 < \dots < p_n\}$, with $p_0 = 0$, we recover the merged block lengths as

$$\delta p_t = p_t - p_{t-1} - 1, \quad t = 1, \dots, n.$$

The placement $Q' = \{q_1 < \dots < q_{n_1}\}$ and its complement $\{\bar{q}_1 < \dots < \bar{q}_{n_2}\}$ then recover the nonterminal block lengths of the two inputs through

$$\delta p_{1,s} = \delta p_{q_s}, \quad \delta p_{2,t} = \delta p_{\bar{q}_t}.$$

Their terminal block lengths follow from

$$\delta p_{l, n_l+1} = m_l - \sum_{s=1}^{n_l} \delta p_{l,s}, \quad l \in \{1, 2\},$$

and hence the defining block-length encoding recovers P_1 and P_2 . Finally, writing $d = \sum_{u=1}^n \delta p_u$, we recover the terminal shuffle placement from

$$R = (Q''_{Q',R}(P_1, P_2) \cap \{d+1, \dots, m\}) - d.$$

Thus, the image determines P_1 , P_2 , Q' , and R uniquely, which proves injectivity. $\square$

For $0 \leq r' < |\text{Pl}(n_1, n_2)|$ and $0 \leq r'' < |\text{Pl}(m_1, m_2)|$, define

$$S_{n_1, m_1, n_2, m_2}^{\bullet, r', r''} \subseteq \{0, \dots, |\text{Pl}(n_1, m_1)| - 1\} \times \{0, \dots, |\text{Pl}(n_2, m_2)| - 1\} \times \{0, \dots, |\text{Pl}(n, m)| - 1\}$$

as the set of all triples

$$(\text{rk}(P_1), \text{rk}(P_2), \text{rk}(P_{Q'}(P_1, P_2)))$$

such that

$$Q' = \text{rk}^{-1}(r'), \quad R \in \text{Pl}(\delta p_{1, n_1+1}, \delta p_{2, n_2+1}), \quad \text{rk}(Q''_{Q',R}(P_1, P_2)) = r''.$$

Equivalently, for $Q' = \text{rk}^{-1}(r')$ and $Q'' = \text{rk}^{-1}(r'')$,

$$S_{n_1, m_1, n_2, m_2}^{\bullet, Q', Q''}(e_{j_1} \otimes e_{j_2}) = \sum_{\substack{0 \leq k < |\text{Pl}(n, m)| \\ (j_1, j_2, k) \in S_{n_1, m_1, n_2, m_2}^{\bullet, r', r''}}} e_k.$$

By [Lemma 4.2](#), no multiplicities or coefficient arrays are required. The lists are generated directly by enumerating P_1 , P_2 , Q' , and R as described in the following algorithm.

Algorithm 3 Placement-rank lists for the ordered shear shuffle**Require:** bidegrees (n_1, m_1) and (n_2, m_2) **Ensure:** the nonempty lists $\mathbf{S}_{n_1, m_1, n_2, m_2}^{\bullet, r', r''}$

```

1:  $n \leftarrow n_1 + n_2, m \leftarrow m_1 + m_2$ 
2: initialize an empty associative collection of lists
3: for all  $P_1 \in \text{Pl}(n_1, m_1)$  with increasing rank  $j_1$  do
4:   compute  $\delta p_{1,1}, \dots, \delta p_{1,n_1+1}$ 
5:   for all  $P_2 \in \text{Pl}(n_2, m_2)$  with increasing rank  $j_2$  do
6:     compute  $\delta p_{2,1}, \dots, \delta p_{2,n_2+1}$ 
7:     for all  $Q' = \{q_1 < \dots < q_{n_1}\} \in \text{Pl}(n_1, n_2)$  with increasing rank  $r'$  do
8:       write  $\{\bar{q}_1 < \dots < \bar{q}_{n_2}\} = \{1, \dots, n\} \setminus Q'$ 
9:       set  $\delta p_{q_s} \leftarrow \delta p_{1,s}$  for  $s = 1, \dots, n_1$ 
10:      set  $\delta p_{\bar{q}_t} \leftarrow \delta p_{2,t}$  for  $t = 1, \dots, n_2$ 
11:       $P \leftarrow P_{Q'}(P_1, P_2)$ 
12:       $k \leftarrow \text{rk}(P)$ 
13:      for all  $R \in \text{Pl}(\delta p_{1,n_1+1}, \delta p_{2,n_2+1})$  with increasing rank do
14:         $Q'' \leftarrow Q'_{Q',R}(P_1, P_2)$ 
15:         $r'' \leftarrow \text{rk}(Q'')$ 
16:        if the key  $(r', r'')$  is absent then
17:          initialize  $\mathbf{S}_{n_1, m_1, n_2, m_2}^{\bullet, r', r''} \leftarrow []$ 
18:        end if
19:        append  $(j_1, j_2, k)$  to  $\mathbf{S}_{n_1, m_1, n_2, m_2}^{\bullet, r', r''}$ 
20:      end for
21:    end for
22:  end for
23: end for
24: return the nonempty keyed lists  $\mathbf{S}_{n_1, m_1, n_2, m_2}^{\bullet, r', r''}$ 

```

Implementation

The nested loops in [Algorithm 3](#) are precomputation only. At runtime, the loops are over the bidegree splittings and the nonempty keys (r', r'') . For one key, write (J_1, J_2, K) for the three stored rank columns. The complete keyed update is

- 1: $\mathbf{V} \leftarrow \text{batchouter}(\ell_1^{(n_1, m_1)}[J_1], \ell_2^{(n_2, m_2)}[J_2])$
- 2: $\mathbf{V} \leftarrow \Theta_{Q', Q''}^{\text{dense}} \mathbf{V}$
- 3: $\mathbf{c}^{(n, m)}[K] += \mathbf{V}$

By [Lemma 4.2](#), the entries of K are distinct within one key. Hence the final line is one vectorized indexed addition and requires no reduction. There is no runtime loop over the stored triples; if a keyed list is too large, it may be processed in contiguous batches. Every term has coefficient one.

4.1.3. Coordinate transformations. Recall from [Section 1.4](#) that Ψ maps standard coordinates to shear coordinates, while Ψ^{-1} maps shear coordinates back to standard coordinates. Both transformations preserve bidegrees and the ordered sequence of $\mathcal{A}'$ -letters. At a fixed bidegree (n, m) , they may therefore change the placement of the $\mathcal{A}'$ -letters and the order of the $\mathcal{A}''$ -letters, but not the order of the $\mathcal{A}'$ -letters.

We first introduce the notation used to describe and store these actions. For $P \in \text{Pl}(n, m)$ and $k = \text{rk}(P)$, recall from [Section 2.2](#) that $\mathbf{a}_k^{(n, m)} \in (\mathbb{R}^{|\mathcal{A}'|})^{\otimes n} \otimes (\mathbb{R}^{|\mathcal{A}''|})^{\otimes m}$ is the placement

slice containing the coefficients of the words whose $\mathcal{A}'$ -letters occupy the positions P . We refer to $\mathbf{a}_\ell^{(n,m)}$ as an input placement slice and to $[\Psi^{\pm 1}(\mathbf{a})^{(n,m)}]_k$ as an output placement slice. Thus, an individual term in either transformation reads one input placement slice, permutes its dense $\mathcal{A}''$ -axes, and adds the resulting tensor to one output placement slice.

We use the following further notation: For tuples $a = (a_1, \dots, a_u)$ and $b = (b_1, \dots, b_v)$, we write

$$a \parallel b := (a_1, \dots, a_u, b_1, \dots, b_v)$$

for their concatenation. Empty tuples and empty index ranges are omitted. We identify a permutation $\sigma \in \mathfrak{S}_m$ with its one-line notation

$$\sigma = (\sigma_1, \dots, \sigma_m).$$

It acts on the dense $\mathcal{A}''$ -axes of a placement slice by

$$\left[(\text{id}^{\otimes n} \otimes \sigma) \mathbf{a}_\ell^{(n,m)} \right]_{i_1, \dots, i_n, j_1, \dots, j_m} = \left[\mathbf{a}_\ell^{(n,m)} \right]_{i_1, \dots, i_n, j_{\sigma_1}, \dots, j_{\sigma_m}}.$$

Thus the dense $\mathcal{A}'$ -axes are left fixed and only the dense $\mathcal{A}''$ -axes are permuted. For constructing permutation tuples, given

$$Q = \{q_1 < \dots < q_u\} \in \text{Pl}(u, v), \quad \{\bar{q}_1 < \dots < \bar{q}_v\} = \{1, \dots, u+v\} \setminus Q,$$

and tuples $a = (a_1, \dots, a_u)$ and $b = (b_1, \dots, b_v)$, we define

$$\text{merge}_Q(a, b) = (c_1, \dots, c_{u+v}), \quad c_{q_s} = a_s, \quad c_{\bar{q}_t} = b_t.$$

Thus $\text{merge}_Q(a, b)$ interleaves the entries of a and b while preserving their respective orders.

The permutation entries in all mathematical formulas are labelled by $1, \dots, m$. By contrast, placement ranks and the integer keys introduced below are indexed from zero. Any conversion of a permutation tuple to the zero-based axis convention of a particular tensor backend is an implementation detail and is not included in the mathematical notation.

At fixed bidegree, we compute each output placement slice as a sparse sum of input placement slices after permuting the dense $\mathcal{A}''$ -axes. By the *support* of either transformation at bidegree (n, m) , we mean the collection of its nonzero contributions, each specified by an output placement, an input placement, and a permutation of the dense $\mathcal{A}''$ -axes. For a graded linear map L and a word $w \in \mathcal{W}_\mathcal{A}$, we write L_w for its graded row, characterized by $\langle w, L(\mathbf{a}) \rangle = \langle L_w, \mathbf{a} \rangle$. The graded-row identities established in [2, Propositions 3.2 and 3.7] provide recursive descriptions of these supports. For $u \in \mathcal{W}_{\mathcal{A}''}$, $w \in \mathcal{W}_\mathcal{A}$, $i \in \mathcal{A}'$, and $i_1, \dots, i_r \in \mathcal{A}''$, these identities are

$$(6) \quad \Psi_u = u, \quad \Psi_{wiu} = (\Psi_w i) \sqcup u,$$

and

$$(7) \quad \Psi_u^{-1} = u, \quad \Psi_{wi_1 \dots i_r}^{-1} = \sum_{q=0}^r (-1)^q (\Psi_w^{-1} \bullet i_q \dots i_1) i_{i_{q+1}} \dots i_r,$$

where empty strings of letters are omitted.

We use the superscript $+$ for the support of Ψ and the superscript $-$ for the support of Ψ^{-1} . For each bidegree (n, m) , let

$$\Sigma_{n,m}^\pm = (\sigma_{n,m,c}^\pm)_{0 \leq c < |\Sigma_{n,m}^\pm|}$$

be the table of the distinct permutations that occur in the corresponding transformation, indexed in order of first occurrence. During construction, a dictionary assigns to each permutation tuple σ its zero-based key

$$\text{key}_{n,m}^\pm(\sigma) \in \{0, \dots, |\Sigma_{n,m}^\pm| - 1\}.$$

If the tuple has already occurred, the dictionary returns its existing key; otherwise, it appends the tuple to $\Sigma_{n,m}^\pm$ and returns the new key.

We encode the sparse support by

$$\mathsf{T}_{n,m}^{\pm} \subseteq \{0, \dots, |\mathsf{Pl}(n, m)| - 1\}^2 \times \{0, \dots, |\Sigma_{n,m}^{\pm}| - 1\}.$$

An entry

$$(k, \ell, c) \in \mathsf{T}_{n,m}^{\pm}$$

means that the input placement slice $\mathbf{a}_{\ell}^{(n,m)}$, after permutation of its dense $\mathcal{A}''$ -axes by $\sigma_{n,m,c}^{\pm}$, contributes to the output placement slice of rank k .

For the recursive construction and the explicit formulas for the transformations, it is convenient to group the support entries by their output rank, since the graded-row recursions determine one output placement slice at a time. We therefore define

$$\mathsf{T}_{n,m}^{\pm}(k) := \{(\ell, c) : (k, \ell, c) \in \mathsf{T}_{n,m}^{\pm}\}.$$

Thus, $\mathsf{T}_{n,m}^{\pm}(k)$ records all pairs consisting of an input placement rank ℓ and a permutation key c whose corresponding term contributes to the output placement slice of rank k .

The recursion for Ψ in (6) has only positive unit coefficients, whereas the recursion for Ψ^{-1} in (7) additionally produces signs. For

$$P = \{p_1 < \dots < p_n\}, \quad G = \{g_1 < \dots < g_n\} \in \mathsf{Pl}(n, m),$$

define the placement parity

$$\eta_{n,m}(\mathsf{rk}(P)) := (-1)^{\sum_{s=1}^n p_s}$$

and

$$\varepsilon_{n,m}(\mathsf{rk}(P), \mathsf{rk}(G)) := (-1)^{\sum_{s=1}^n (g_s - p_s)} = \eta_{n,m}(\mathsf{rk}(P))\eta_{n,m}(\mathsf{rk}(G)).$$

Lemma 4.3. *For every bidegree (n, m) , the permutation tables and support relations produced by Algorithms 4 and 5 represent the transformations as follows. For every $\mathbf{a} \in T^{N,M}(\mathbb{R}^d)$ and every $0 \leq k < |\mathsf{Pl}(n, m)|$,*

$$[\Psi(\mathbf{a})^{(n,m)}]_k = \sum_{(\ell, c) \in \mathsf{T}_{n,m}^+(k)} (\mathrm{id}^{\otimes n} \otimes \sigma_{n,m,c}^+) \mathbf{a}_{\ell}^{(n,m)}$$

and

$$[\Psi^{-1}(\mathbf{a})^{(n,m)}]_k = \sum_{(\ell, c) \in \mathsf{T}_{n,m}^-(k)} \varepsilon_{n,m}(k, \ell) (\mathrm{id}^{\otimes n} \otimes \sigma_{n,m,c}^-) \mathbf{a}_{\ell}^{(n,m)}.$$

The pairs (ℓ, c) in each fixed-output fibre are pairwise distinct. In particular, every term in the formula for Ψ has coefficient one, while the coefficient of a term in the formula for Ψ^{-1} is determined by its input and output placement ranks through $\varepsilon_{n,m}$.

The support tables in Lemma 4.3 are constructed recursively in the $\mathcal{A}'$ -degree. For $n = 0$, both transformations are the identity on words over $\mathcal{A}''$, and we initialize

$$\Sigma_{0,m}^+ = \Sigma_{0,m}^- = ((1, \dots, m)), \quad \mathsf{T}_{0,m}^+ = \mathsf{T}_{0,m}^- = \{(0, 0, 0)\}.$$

Let $n \geq 1$ and $P = \{p_1 < \dots < p_n\} \in \mathsf{Pl}(n, m)$. Every basis word of bidegree (n, m) with $\mathcal{A}'$ -placement P has a unique factorization

$$w = v \overset{\text{blue}}{i} \overset{\text{blue}}{i}_1 \cdots \overset{\text{blue}}{i}_{n+m-p_n},$$

where v has bidegree $(n-1, p_n - n)$ and placement

$$P^- := \{p_1, \dots, p_{n-1}\} \in \mathsf{Pl}(n-1, p_n - n).$$

Its rank satisfies

$$\mathsf{rk}(P^-) = \mathsf{rk}(P) - \binom{p_n - 1}{n}.$$

For Ψ , (6) first appends $\text{blue } i$ to every parent term represented by

$$\mathsf{T}_{n-1, p_n - n}^+(\mathsf{rk}(P^-)).$$

If (ℓ, c) belongs to this fibre, then the appended parent placement has rank

$$\ell + \binom{p_n - 1}{n}$$

in $\text{Pl}(n, p_n - n)$. It is then shuffled with the terminal block of $n + m - p_n$ $\mathcal{A}''$ -letters. This is the standard shuffle of bidegrees $(n, p_n - n)$ and $(0, n + m - p_n)$, so its placement ranks are read from the precomputed lists

$$S_{n, p_n - n, 0, n + m - p_n}^{0, r''}.$$

Each such list is stored as two input-rank columns in increasing implicit output rank. During the construction of the transformation supports, we use a temporary reverse index from an input-rank pair to the zero-based row positions at which it occurs.

Algorithm 4 Sparse support for Ψ

Require: a bidegree (n, m) , the supports and permutation tables of lower $\mathcal{A}'$ -degree, the standard shuffle lists, and their temporary reverse indices

Ensure: the permutation table $\Sigma_{n,m}^+$ and the support $\mathsf{T}_{n,m}^+$

```

1: if  $n = 0$  then
2:    $\Sigma_{0,m}^+ \leftarrow [(1, \dots, m)]$ 
3:    $\mathsf{T}_{0,m}^+ \leftarrow [(0, 0, 0)]$ 
4: else
5:    $\Sigma_{n,m}^+ \leftarrow []$ 
6:   initialize an empty dictionary from permutation tuples to keys
7:    $\mathsf{T}_{n,m}^+ \leftarrow []$ 
8:   for all  $P = \{p_1 < \dots < p_n\} \in \text{Pl}(n, m)$  with increasing rank  $k$  do
9:      $P^- \leftarrow \{p_1, \dots, p_{n-1}\}$ 
10:    for all  $(\ell, c) \in \mathsf{T}_{n-1, p_n - n}^+(\text{rk}(P^-))$  do
11:       $\sigma \leftarrow \sigma_{n-1, p_n - n, c}^+$ 
12:      for all  $Q'' \in \text{Pl}(p_n - n, n + m - p_n)$  with increasing rank  $r''$  do
13:        for all zero-based row positions  $h$  at which the two stored columns of
           $S_{n, p_n - n, 0, n + m - p_n}^{0, r''}$  equal  $(\ell + \binom{p_n - 1}{n}, 0)$  do
14:           $\tilde{\sigma} \leftarrow \text{merge}_{Q''}((\sigma_1, \dots, \sigma_{p_n - n}), (p_n - n + 1, \dots, m))$ 
15:           $\tilde{c} \leftarrow \text{key}_{n,m}^+(\tilde{\sigma})$ 
16:          append  $(k, h, \tilde{c})$  to  $\mathsf{T}_{n,m}^+$ 
17:        end for
18:      end for
19:    end for
20:  end for
21: end if
22: return  $\Sigma_{n,m}^+$  and  $\mathsf{T}_{n,m}^+$ 

```

For Ψ^{-1} , (7) starts from

$$\mathsf{T}_{n-1, p_n - n}^-(\text{rk}(P^-)).$$

For (ℓ, c) in this fibre, write

$$H = \text{rk}^{-1}(\ell) \in \text{Pl}(n - 1, p_n - n).$$

The length of the terminal $\mathcal{A}''$ -block of the corresponding parent term is

$$\delta h_n = \begin{cases} p_n - n, & n = 1, \\ p_n - 1 - h_{n-1}, & n \geq 2, \end{cases}$$

where $H = \{h_1 < \dots < h_{n-1}\}$ in the second case. Thus, the final δh_n entries of $\sigma_{n-1, p_n-n, c}^-$ index the terminal block of the parent term.

For $0 \leq q \leq n + m - p_n$, the reversed labels

$$(p_n - n + q, \dots, p_n - n + 1)$$

are interleaved with this terminal block, the new $\mathcal{A}'$ -letter is appended, and the remaining labels $p_n - n + q + 1, \dots, m$ stay after it. The resulting input placement is

$$H \cup \{p_n + q\} \in \text{Pl}(n, m)$$

and has rank

$$\ell + \binom{p_n + q - 1}{n}.$$

Proof of Lemma 4.3. We prove the two assertions simultaneously by induction on the $\mathcal{A}'$ -degree n . By linearity and the definition of the graded rows, it is enough to show that, for each basis word with a fixed output placement, the corresponding algorithm generates exactly the terms of its graded-row expansion, with the indicated input placement, dense-axis permutation, and coefficient.

For $n = 0$, both transformations are the identity on words over $\mathcal{A}''$. The initialization

$$\Sigma_{0,m}^+ = \Sigma_{0,m}^- = ((1, \dots, m)), \quad \mathsf{T}_{0,m}^+ = \mathsf{T}_{0,m}^- = \{(0, 0, 0)\}$$

therefore gives both formulas.

Suppose that the assertions hold for $\mathcal{A}'$ -degree $n - 1$, and fix

$$P = \{p_1 < \dots < p_n\} \in \text{Pl}(n, m).$$

Every row word with placement P has a unique factorization

$$w = v \textcolor{blue}{i} u, \quad u \in \mathcal{W}_{\mathcal{A}''}, \quad |u| = n + m - p_n,$$

where v has placement P^- .

For Ψ , the row recursion

$$\Psi_{v \textcolor{blue}{i} u} = (\Psi_v \textcolor{blue}{i}) \sqcup u$$

first appends $\textcolor{blue}{i}$ to each term of Ψ_v and then shuffles the result with the terminal word u . By the induction hypothesis, the parent terms are represented exactly once by

$$(\ell, c) \in \mathsf{T}_{n-1, p_n-n}^+(\text{rk}(P^-)).$$

Appending $\textcolor{blue}{i}$ changes the corresponding input placement rank to

$$\ell + \binom{p_n - 1}{n}.$$

The subsequent ordinary shuffle is enumerated by the precomputed standard shuffle lists. For each

$$Q'' \in \text{Pl}(p_n - n, n + m - p_n),$$

the induced permutation of the $\mathcal{A}''$ -axes is

$$\text{merge}_{Q''}((\sigma_1, \dots, \sigma_{p_n-n}), (p_n - n + 1, \dots, m)),$$

which is exactly the tuple constructed in Algorithm 4. Hence the algorithm generates every term prescribed by (6), and every entry generated by the algorithm arises from one such term.

To prove uniqueness, consider a pair $(h, \tilde{c})$ generated for the fixed output placement P . The key $\tilde{c}$ determines the tuple $\sigma_{n,m,\tilde{c}}^+$. The entries with labels $1, \dots, p_n - n$ determine the placement Q'' within this tuple, and reading these entries in order recovers the parent permutation. The rank h , together with Q'' and the fixed second input placement, recovers the first input placement of the ordinary shuffle by Lemma 3.3. Subtracting $\binom{p_n-1}{n}$ from its rank recovers the parent input rank, and the parent permutation table then recovers its key. The parent pair is unique by the induction

Algorithm 5 Sparse support for Ψ^{-1} **Require:** a bidegree (n, m) and the supports and permutation tables of lower $\mathcal{A}'$ -degree**Ensure:** the permutation table $\Sigma_{n,m}^-$ and the support $\mathsf{T}_{n,m}^-$

```

1: if  $n = 0$  then
2:    $\Sigma_{0,m}^- \leftarrow [(1, \dots, m)]$ 
3:    $\mathsf{T}_{0,m}^- \leftarrow [(0, 0, 0)]$ 
4: else
5:    $\Sigma_{n,m}^- \leftarrow []$ 
6:   initialize an empty dictionary from permutation tuples to keys
7:    $\mathsf{T}_{n,m}^- \leftarrow []$ 
8:   for all  $P = \{p_1 < \dots < p_n\} \in \text{Pl}(n, m)$  with increasing rank  $k$  do
9:      $P^- \leftarrow \{p_1, \dots, p_{n-1}\}$ 
10:    for all  $(\ell, c) \in \mathsf{T}_{n-1, p_n-n}^-(\text{rk}(P^-))$  do
11:       $\sigma \leftarrow \sigma_{n-1, p_n-n, c}^-$ 
12:       $H \leftarrow \text{rk}^{-1}(\ell)$ 
13:      if  $n = 1$  then
14:         $\delta h_n \leftarrow p_n - n$ 
15:      else
16:        write  $H = \{h_1 < \dots < h_{n-1}\}$ 
17:         $\delta h_n \leftarrow p_n - 1 - h_{n-1}$ 
18:      end if
19:      for  $q = 0, \dots, n + m - p_n$  do
20:        for all  $Q'' \in \text{Pl}(\delta h_n, q)$  with increasing rank do
21:           $\tilde{\sigma} \leftarrow (\sigma_1, \dots, \sigma_{p_n-n-\delta h_n}) \parallel$ 
            merge $_{Q''}((\sigma_{p_n-n-\delta h_n+1}, \dots, \sigma_{p_n-n}), (p_n - n + q, \dots, p_n - n + 1)) \parallel$ 
             $(p_n - n + q + 1, \dots, m)$ 
22:           $\tilde{c} \leftarrow \text{key}_{n,m}^-(\tilde{\sigma})$ 
23:          append
            
$$\left(k, \ell + \binom{p_n + q - 1}{n}, \tilde{c}\right)$$

        end for
      end for
    end for
  end if
29: return  $\Sigma_{n,m}^-$  and  $\mathsf{T}_{n,m}^-$ 

```

hypothesis. Thus, the pairs in $\mathsf{T}_{n,m}^+(\text{rk}(P))$ are pairwise distinct. Finally, every ordinary-shuffle term occurs once with coefficient one, so the asserted formula for Ψ follows.

For Ψ^{-1} , write the row word as

$$w = v \mathbf{i} i_1 \cdots i_r, \quad r = n + m - p_n.$$

The recursion

$$\Psi_{v \mathbf{i} i_1 \cdots i_r}^{-1} = \sum_{q=0}^r (-1)^q (\Psi_v^{-1} \bullet i_q \cdots i_1) \mathbf{i} i_{q+1} \cdots i_r$$

starts from the parent terms represented by

$$\mathsf{T}_{n-1, p_n-n}^-(\text{rk}(P^-)).$$

For a parent input placement

$$H = \{h_1 < \dots < h_{n-1}\}$$

and a fixed q , the product with the pure $\mathcal{A}''$ -word $i_q \dots i_1$ shuffles that word with the terminal $\mathcal{A}''$ -block of the parent term. These shuffles are indexed by

$$Q'' \in \text{Pl}(\delta h_n, q).$$

The resulting input placement is

$$H \cup \{p_n + q\},$$

whose rank is

$$\text{rk}(H) + \binom{p_n + q - 1}{n},$$

and the corresponding dense-axis permutation is precisely the tuple $\tilde{\sigma}$ constructed in [Algorithm 5](#). Thus, the algorithm generates exactly the terms prescribed by (7).

It remains to prove uniqueness and identify the sign. Consider a generated pair $(h, \tilde{c})$ and write

$$G = \text{rk}^{-1}(h) = \{g_1 < \dots < g_n\}.$$

At the final recursive step, the newly appended $\mathcal{A}'$ -letter occupies position

$$g_n = p_n + q.$$

Hence,

$$q = g_n - p_n$$

is determined by the input and output placements. Removing g_n recovers the parent input placement

$$H = \{g_1, \dots, g_{n-1}\}$$

and therefore its terminal block length δh_n .

The key $\tilde{c}$ determines the tuple $\sigma_{n,m,\tilde{c}}^-$. Its middle interleaving window consists of the entries in positions

$$p_n - n - \delta h_n + 1, \dots, p_n - n + q.$$

Within this window, the entries with labels at most $p_n - n$ determine Q'' , and these entries read in order, together with the unchanged prefix, recover the parent permutation. The parent permutation table then recovers its key. The parent pair is unique by the induction hypothesis, and hence the pairs in $\mathbb{T}_{n,m}^-(\text{rk}(P))$ are pairwise distinct.

At the recursive step with s $\mathcal{A}'$ -letters, the selected index is $g_s - p_s$. Therefore, the product of the signs generated by the recursion is

$$\prod_{s=1}^n (-1)^{g_s - p_s} = (-1)^{\sum_{s=1}^n (g_s - p_s)} = \varepsilon_{n,m}(\text{rk}(P), \text{rk}(G)).$$

This proves the asserted formula for Ψ^{-1} and completes the induction. $\square$

Implementation

The nested loops in [Algorithms 4](#) and [5](#) are precomputation only. During this construction, keep the support grouped by output rank so that the parent fibres required at the next $\mathcal{A}'$ -degree are contiguous. Afterward, regroup the same entries by permutation key c and retain their output- and input-rank lists K_c and L_c . For the inverse, the sign of a retained pair (k, ℓ) is $\eta_{n,m}(k)\eta_{n,m}(\ell)$. For a small group for which it is memory-competitive, one may additionally materialize the corresponding dense rank matrix M_c .

At runtime, the loops are over the bidegrees and the permutation keys only. For one key, apply

- 1: **if** a dense rank matrix M_c is retained **then**
- 2: $\mathbf{X} \leftarrow \text{reshape}((\text{id}^{\otimes n} \otimes \sigma_c) \mathbf{a}^{(n,m)})$

```

3:    $\mathbf{Y} += M_c \mathbf{X}$ 
4: else
5:    $\mathbf{V} \leftarrow \mathbf{a}^{(n,m)}[L_c]$ 
6:   apply  $\sigma_c$  to the dense  $\mathcal{A}''$ -axes of  $\mathbf{V}$ 
7:   flatten the dense axes of  $\mathbf{V}$  into one trailing axis
8:   if applying  $\Psi^{-1}$  then
9:     multiply each row of  $\mathbf{V}$  by  $\eta_{n,m}(K_c)\eta_{n,m}(L_c)$ 
10:  end if
11:   $\mathbf{Y} \leftarrow \text{scatteradd}(\mathbf{Y}, K_c, \mathbf{V})$ 
12: end if

```

Thus there is no runtime loop over support entries, placement ranks, or dense coordinates. Dense groups permute the full block and use one rank-axis matrix multiplication. All other groups first gather the required placement slices, then permute only this batch and scatter-add it along the rank axis. In particular, the latter are retained as rank-pair lists rather than as sparse matrices.

Finally, let $P = \{p_1 < \dots < p_n\}$ and $G = \{g_1 < \dots < g_n\}$ be the output and input placements of a support entry. Then

$$g_s \geq p_s, \quad s = 1, \dots, n.$$

For Ψ , this follows because each ordinary shuffle moves the $\mathcal{A}'$ -letters only to the right, while for Ψ^{-1} it follows from the nonnegative peeling indices $g_s - p_s$. Consequently, both transformations are upper triangular in colexicographic order. Equality of the two placements forces every shuffle displacement, respectively every peeling index, to vanish, so the diagonal contains only the identity dense-axis permutation. All transformation data depend only on the truncation parameters.

With the graded-transpose convention of [2], transposing the precomputed block maps for Ψ gives Ψ^* . Hence a functional ℓ in shear coordinates is paired with a tensor $\mathbf{a}$ in standard coordinates as $\langle \ell, \mathbf{a} \rangle_\Psi = \langle \Psi^*(\ell), \mathbf{a} \rangle$, without first forming $\Psi(\mathbf{a})$.

4.1.4. Computing signatures. Since Ψ preserves bidegrees, $\pi^{N,M} \circ \Psi = \Psi \circ \pi^{N,M}$. Thus, one may compute the standard bidegree-truncated signature by Algorithm 2 and apply the transformation Ψ from Section 4.1.3 once at the end. This is the natural procedure when only the final shear coordinates of the signature are required.

There is also a direct shear-coordinate recursion. Although a general product $\mathbf{a} \odot_{N,M} \mathbf{b}$ is evaluated through the coordinate transformations, the Horner recursion only requires right multiplication by a first-level tensor $z = (z', z'') \in \mathbb{R}^d$.

We first describe this specialized action. Let $n > 0$ and $P = \{p_1 < \dots < p_n\} \in \text{Pl}(n, m)$. We set $p_0 = 0$ and

$$P^- := \{p_1, \dots, p_{n-1}\} \in \text{Pl}(n-1, m).$$

For $Q \in \text{Pl}(p_n - p_{n-1} - 1, n + m - p_n)$, define

$$\rho_{P,Q} := (1, \dots, p_{n-1} - n + 1) \parallel \text{merge}_Q((p_{n-1} - n + 2, \dots, p_n - n), (p_n - n + 1, \dots, m)) \in \mathfrak{S}_m,$$

where empty tuples are omitted. Thus, $\rho_{P,Q}$ leaves unchanged the $\mathcal{A}''$ -axes preceding the last two blocks and interleaves the two blocks adjacent to the final $\mathcal{A}'$ -letter according to Q .

Lemma 4.4. *Let $\mathbf{a} \in T^{N,M}(\mathbb{R}^d)$ and $z = (z', z'') \in \mathbb{R}^d$, and set $\mathbf{c} := \pi^{N,M}(\mathbf{a} \odot z)$. Fix $(n, m) \in \{0, \dots, N\} \times \{0, \dots, M\}$ and $0 \leq k < |\text{Pl}(n, m)|$. If $n > 0$, write $P = \text{rk}^{-1}(k) = \{p_1 < \dots < p_n\}$,*

set $p_0 = 0$, and let $P^- = \{p_1, \dots, p_{n-1}\}$. Then

$$\mathbf{c}_k^{(n,m)} = \begin{cases} \mathbf{a}_k^{(n,m-1)} \otimes z'', & m > 0 \text{ and } k < |\text{Pl}(n, m-1)|, \\ 0, & \text{otherwise,} \end{cases}$$

$$+ \begin{cases} \sum_{Q \in \text{Pl}(p_n - p_{n-1} - 1, n + m - p_n)} \tau_{n-1, m, 1, 0} \left((\text{id}^{\otimes n-1} \otimes \rho_{P, Q}) \mathbf{a}_{\text{rk}(P^-)}^{(n-1, m)} \otimes z' \right), & n > 0, \\ 0, & n = 0. \end{cases}$$

Proof. Since Ψ and Ψ^{-1} are the identity on the first level, the definition of the transported product gives, for every basis word w ,

$$\langle w, \mathbf{a} \odot z \rangle = \langle \Psi_w, \Psi^{-1}(\mathbf{a}) \otimes z \rangle.$$

Consider first the z'' -contribution. If a word w ends in the letter $j \in \mathcal{A}''$, then the terms of Ψ_w ending in j sum to $\Psi_{\tilde{w}}j$, where $\tilde{w}$ is obtained by removing the final letter of w . Hence the corresponding contribution is $\langle \tilde{w}, \mathbf{a} \rangle z''^j$. Removing a final $\mathcal{A}''$ -letter leaves the placement unchanged. The placements of such words form the natural copy of $\text{Pl}(n, m-1)$ inside $\text{Pl}(n, m)$ and occupy precisely the ranks $0, \dots, |\text{Pl}(n, m-1)| - 1$. This proves the first formula.

Now let $n > 0$ and write a word with placement P as

$$w = w_0 i_1 w_1 \cdots w_{n-1} i_n w_n, \quad w_0, \dots, w_n \in \mathcal{W}_{\mathcal{A}''}.$$

If $v = w_0 i_1 w_1 \cdots w_{n-1}$, then the row recursion gives $\Psi_w = (\Psi_v i_n) \sqcup w_n$. The terms in this shuffle ending in i_n sum to $(\Psi_v \sqcup w_n) i_n$. Applying the row recursion once more to v and using associativity of the shuffle product shows that these are exactly the rows obtained by removing i_n and replacing the two adjacent blocks w_{n-1} and w_n by the terms of $w_{n-1} \sqcup w_n$. For $n = 1$, the same conclusion follows from $\Psi_{w_0} = w_0$.

The lengths of these two blocks are $p_n - p_{n-1} - 1$ and $n + m - p_n$, respectively. Their shuffle terms are therefore indexed by $Q \in \text{Pl}(p_n - p_{n-1} - 1, n + m - p_n)$. The blocks preceding them are unchanged, and the induced permutation of the dense $\mathcal{A}''$ -axes is precisely $\rho_{P, Q}$. Removing the final $\mathcal{A}'$ -letter changes the placement from P to P^- , while $\tau_{n-1, m, 1, 0}$ restores the ordered dense-axis layout after tensoring with z' . Summing over Q proves the second formula. $\square$

Since Ψ is an algebra isomorphism from $(T((\mathbb{R}^d)), \otimes)$ to $(T((\mathbb{R}^d)), \odot)$, we define

$$\exp_{\odot}(z) := \Psi(\exp_{\otimes}(z)) = \sum_{r=0}^{\infty} \frac{z^{\odot r}}{r!},$$

with $z^{\odot 0} = e_{\emptyset}$. The bidegree-truncated product $\pi^{N, M}(\mathbf{a} \odot \exp_{\odot}(z))$ is computed by the outer recursion of [Algorithm 2](#), using the first-level action from [Lemma 4.4](#).

Implementation

The runtime loops over q , the active bidegrees, and the permutation keys c . There is no loop over the pairs (P, Q) . For the z' -contribution, regroup the precomputed triples by c and write K_c and L_c for their output- and input-rank columns. For fixed (q, n, m) , use

```

1:  $\mathbf{b}_{q-1}^{(n,m)} \leftarrow \mathbf{a}^{(n,m)}$ 
2: if  $m > 0$  then
3:    $\mathbf{b}_{q-1}^{(n,m)}[0 : |\text{Pl}(n, m-1)|] += q^{-1} \mathbf{b}_q^{(n,m-1)} \otimes z''$ 
4: end if
5: if  $n > 0$  then
6:   for all permutation keys  $c$  do
7:      $\mathbf{V} \leftarrow \mathbf{b}_q^{(n-1,m)}[L_c]$ 

```

```

8:       $\mathbf{V} \leftarrow (\text{id}^{\otimes n-1} \otimes \sigma_c) \mathbf{V}$ 
9:       $\mathbf{V} \leftarrow \tau_{n-1,m,1,0}(\mathbf{V} \otimes z')$ 
10:      $\mathbf{b}_{q-1}^{(n,m)}[K_c] += q^{-1} \mathbf{V}$ 
11:   end for
12: end if

```

For fixed c , the entries of K_c are distinct, so every inner update is vectorized over the whole keyed rank list and all dense axes, without a reduction. Only two Horner states are stored, and neither a general $\odot_{N,M}$ -product nor a coordinate transformation is evaluated inside the update.

The additional work in the direct recursion is confined to shuffling the two $\mathcal{A}''$ -blocks adjacent to the final $\mathcal{A}'$ -letter.

4.2. Partially symmetrized coordinates.

4.2.1. *Concatenation.* Since $\widehat{\Psi}$ and $\widehat{\Psi}^{-1}$ preserve bidegrees, the transported product from [Section 1.6](#) restricts to

$$\widehat{\mathbf{a}} \widehat{\odot}_{N,M} \widehat{\mathbf{b}} = \widehat{\Psi} \left(\widehat{\Psi}^{-1}(\widehat{\mathbf{a}}) \widehat{\odot}_{N,M} \widehat{\Psi}^{-1}(\widehat{\mathbf{b}}) \right), \quad \widehat{\mathbf{a}}, \widehat{\mathbf{b}} \in \widehat{T}^{N,M}(\mathbb{R}^d).$$

4.2.2. *Shuffle.* Since $\widehat{\bullet}$ adds bidegrees, it induces a truncated product

$$\widehat{\bullet}_{N,M} : \mathbb{R}_{\mathcal{A}''}^{N,M} \langle \mathcal{A} \rangle \times \mathbb{R}_{\mathcal{A}''}^{N,M} \langle \mathcal{A} \rangle \longrightarrow \mathbb{R}_{\mathcal{A}''}^{N,M} \langle \mathcal{A} \rangle.$$

Recall from [\[2, Theorem 3.10\]](#) that this product interleaves the two sequences of $\mathcal{A}''$ -block/ $\mathcal{A}'$ -letter units, while the two terminal $\mathcal{A}''$ -blocks are added and contribute a multi-binomial coefficient.

Fix bidegrees (n_1, m_1) and (n_2, m_2) , and write $n = n_1 + n_2$ and $m = m_1 + m_2$. Let

$$\boldsymbol{\alpha} = (\alpha_0, \dots, \alpha_{n_1}) \in \text{MPl}(n_1, m_1), \quad \boldsymbol{\beta} = (\beta_0, \dots, \beta_{n_2}) \in \text{MPl}(n_2, m_2).$$

For $Q' \in \text{Pl}(n_1, n_2)$, define

$$\gamma_{Q'}(\boldsymbol{\alpha}, \boldsymbol{\beta}) := \text{merge}_{Q'}((\alpha_0, \dots, \alpha_{n_1-1}), (\beta_0, \dots, \beta_{n_2-1})) \parallel (\alpha_{n_1} + \beta_{n_2}) \in \text{MPl}(n, m),$$

where empty tuples are omitted. The corresponding coefficient is

$$\lambda(\boldsymbol{\alpha}, \boldsymbol{\beta}) := \prod_{i \in \mathcal{A}''} \binom{\alpha_{n_1,i} + \beta_{n_2,i}}{\alpha_{n_1,i}}.$$

Thus, the nonterminal blocks remain attached to their respective $\mathcal{A}'$ -letters, while only the two terminal blocks are merged.

Let $\widehat{\Theta}_{Q'}$ be the canonical permutation which moves the two multiset-placement axes to the front and places the n_1 dense $\mathcal{A}'$ -axes of the first input in the positions Q' , with the axes of the second input in the complementary positions. For fixed Q' , define the rank-axis map $\widehat{S}_{n_1, m_1, n_2, m_2}^{\bullet, Q'}$ on basis elements by

$$\begin{aligned} & \widehat{S}_{n_1, m_1, n_2, m_2}^{\bullet, Q'} (e_{\text{rk}_{\text{sym}}}(\boldsymbol{\alpha}) \otimes e_{\text{rk}_{\text{sym}}}(\boldsymbol{\beta})) \\ & := \lambda(\boldsymbol{\alpha}, \boldsymbol{\beta}) e_{\text{rk}_{\text{sym}}}(\gamma_{Q'}(\boldsymbol{\alpha}, \boldsymbol{\beta})). \end{aligned}$$

Lemma 4.5. For $\ell_1, \ell_2 \in \mathbb{R}_{\mathcal{A}''}^{N,M} \langle \mathcal{A} \rangle$,

$$(\ell_1 \widehat{\bullet}_{N,M} \ell_2)^{(n,m)} = \sum_{\substack{n_1+n_2=n \\ m_1+m_2=m}} \sum_{Q' \in \text{Pl}(n_1, n_2)} \left(\widehat{S}_{n_1, m_1, n_2, m_2}^{\bullet, Q'} \otimes \text{id} \right) \widehat{\Theta}_{Q'} \left(\ell_1^{(n_1, m_1)} \otimes \ell_2^{(n_2, m_2)} \right).$$

Proof. By bilinearity, it suffices to consider two basis elements in normal form. The recursion for $\hat{\bullet}$ interleaves the two ordered sequences of $\mathcal{A}''$ -block/ $\mathcal{A}'$ -letter units. These interleavings are indexed by $Q' \in \text{Pl}(n_1, n_2)$, and the nonterminal blocks remain attached to their respective $\mathcal{A}'$ -letters. The resulting multiset placement is therefore $\gamma_{Q'}(\alpha, \beta)$, while the dense $\mathcal{A}'$ -axes are interleaved by $\hat{\Theta}_{Q'}$.

The two terminal blocks are combined only after all units have been interleaved. The first identity in the recursion for $\hat{\bullet}$ replaces them by $\alpha_{n_1} + \beta_{n_2}$ and contributes $\lambda(\alpha, \beta)$. This coefficient is independent of Q' . Every interleaving occurs exactly once, so summing over Q' and over all bidegree splittings gives the stated formula. $\square$

Let $r' = \text{rk}(Q')$ and define the multiset-placement rank list

$$\hat{\mathbf{S}}_{n_1, m_1, n_2, m_2}^{\bullet, r'} := \left\{ (\text{rk}_{\text{sym}}(\alpha), \text{rk}_{\text{sym}}(\beta), \text{rk}_{\text{sym}}(\gamma_{Q'}(\alpha, \beta))) : \alpha \in \text{MPI}(n_1, m_1), \right. \\ \left. \beta \in \text{MPI}(n_2, m_2) \right\}.$$

Since the coefficient does not depend on Q' , define it once for each pair of input ranks by

$$\hat{\lambda}_{n_1, m_1, n_2, m_2}(\text{rk}_{\text{sym}}(\alpha), \text{rk}_{\text{sym}}(\beta)) := \lambda(\alpha, \beta).$$

Algorithm 6 Multiset-placement rank lists for the partially symmetrized shear shuffle

Require: bidegrees (n_1, m_1) and (n_2, m_2)

Ensure: the lists $\hat{\mathbf{S}}_{n_1, m_1, n_2, m_2}^{\bullet, r'}$ and the coefficient array $\hat{\lambda}_{n_1, m_1, n_2, m_2}$

- 1: $n \leftarrow n_1 + n_2, m \leftarrow m_1 + m_2$
 - 2: initialize the lists $\hat{\mathbf{S}}_{n_1, m_1, n_2, m_2}^{\bullet, r'}$ as empty
 - 3: **for all** $\alpha = (\alpha_0, \dots, \alpha_{n_1}) \in \text{MPI}(n_1, m_1)$ with increasing rank j_1 **do**
 - 4: **for all** $\beta = (\beta_0, \dots, \beta_{n_2}) \in \text{MPI}(n_2, m_2)$ with increasing rank j_2 **do**
 - 5: $\hat{\lambda}_{n_1, m_1, n_2, m_2}(j_1, j_2) \leftarrow \prod_{i \in \mathcal{A}''} \binom{\alpha_{n_1, i} + \beta_{n_2, i}}{\alpha_{n_1, i}}$
 - 6: **for all** $Q' \in \text{Pl}(n_1, n_2)$ with increasing rank r' **do**
 - 7: $\gamma \leftarrow \text{merge}_{Q'}((\alpha_0, \dots, \alpha_{n_1-1}), (\beta_0, \dots, \beta_{n_2-1})) \parallel (\alpha_{n_1} + \beta_{n_2})$
 - 8: $k \leftarrow \text{rk}_{\text{sym}}(\gamma)$
 - 9: append (j_1, j_2, k) to $\hat{\mathbf{S}}_{n_1, m_1, n_2, m_2}^{\bullet, r'}$
 - 10: **end for**
 - 11: **end for**
 - 12: **end for**
 - 13: **return** the rank lists and the coefficient array
-

Implementation

The nested rank loops in [Algorithm 6](#) are precomputation only. At runtime, the loops are over the bidegree splittings and $Q' \in \text{Pl}(n_1, n_2)$. For one Q' , vectorize over the full input-pair axis and store its target-rank array $K_{Q'}$ in lexicographic input-pair order. The coefficient array $\hat{\lambda}$ is common to all Q' . The keyed update is

- 1: $\mathbf{V} \leftarrow \hat{\Theta}_{Q'}(\ell_1^{(n_1, m_1)} \otimes \ell_2^{(n_2, m_2)})$
- 2: flatten the two rank axes of $\mathbf{V}$
- 3: $\mathbf{V} \leftarrow \hat{\lambda} \mathbf{V}$
- 4: $\hat{\mathbf{c}}^{(n, m)} \leftarrow \text{scatteradd}(\hat{\mathbf{c}}^{(n, m)}, K_{Q'}, \mathbf{V})$

Thus there is no runtime loop over (j_1, j_2) . Distinct input pairs that have the same output rank are accumulated by the scatter-add itself; no output-rank ordering, segment offsets, or per-segment target array is stored. If needed, the flattened pair axis may be processed in batches with updates scattered to the same target ranks.

4.2.3. Coordinate transformations. The transformations $\widehat{\Psi}$ and $\widehat{\Psi}^{-1}$ preserve bidegrees and the ordered sequence of $\mathcal{A}'$ -letters. Since the partially symmetrized block layout has no dense $\mathcal{A}''$ -axes, their action at bidegree (n, m) reduces to a sparse integer matrix on the multiset-placement axis. We use the superscript $+$ for $\widehat{\Psi}$ and the superscript $-$ for $\widehat{\Psi}^{-1}$.

For $\alpha = (\alpha_0, \dots, \alpha_n) \in \text{MPI}(n, m)$, let

$$P(\alpha) := \left\{ s + \sum_{r=0}^{s-1} |\alpha_r| : s = 1, \dots, n \right\}$$

be its induced ordered placement and define

$$\widehat{\eta}_{n,m}(\text{rk}_{\text{sym}}(\alpha)) := (-1)^{\sum_{s=1}^n (s + \sum_{r=0}^{s-1} |\alpha_r|)}.$$

For two multiset-placement ranks k and ℓ , set

$$\widehat{\varepsilon}_{n,m}(k, \ell) := \widehat{\eta}_{n,m}(k) \widehat{\eta}_{n,m}(\ell).$$

Lemma 4.6. *For every bidegree (n, m) , [Algorithms 7](#) and [8](#) produce sparse matrices*

$$\widehat{C}_{n,m}^{\pm} \in \mathbb{N}_0^{|\text{MPI}(n,m)| \times |\text{MPI}(n,m)|}$$

such that, for every $\widehat{\mathbf{a}} \in \widehat{T}^{N,M}(\mathbb{R}^d)$ and every $0 \leq k < |\text{MPI}(n, m)|$,

$$\left[\widehat{\Psi}(\widehat{\mathbf{a}})^{(n,m)} \right]_k = \sum_{\ell} \widehat{C}_{n,m}^+(k, \ell) \widehat{\mathbf{a}}_{\ell}^{(n,m)}$$

and

$$\left[\widehat{\Psi}^{-1}(\widehat{\mathbf{a}})^{(n,m)} \right]_k = \sum_{\ell} \widehat{\varepsilon}_{n,m}(k, \ell) \widehat{C}_{n,m}^-(k, \ell) \widehat{\mathbf{a}}_{\ell}^{(n,m)}.$$

We construct the matrices in increasing $\mathcal{A}'$ -degree. For $n = 0$, both transformations are the identity, so

$$\widehat{C}_{0,m}^+ = \widehat{C}_{0,m}^- = I_{|\text{MPI}(0,m)|}.$$

Let $n \geq 1$ and fix

$$\alpha = (\alpha_0, \dots, \alpha_n) \in \text{MPI}(n, m).$$

Writing $\varrho = |\alpha_n|$, the parent output multiset placement is

$$\alpha^- := (\alpha_0, \dots, \alpha_{n-1}) \in \text{MPI}(n-1, m-\varrho).$$

For $\widehat{\Psi}$, the terminal block α_n is shuffled with each term of the parent row. In the quotient, this amounts to distributing α_n among the $n+1$ blocks of the appended parent term. Thus, for each componentwise decomposition

$$\alpha_n = \nu_0 + \dots + \nu_n,$$

a parent input multiset placement $(\gamma_0, \dots, \gamma_{n-1})$ contributes to

$$(\gamma_0 + \nu_0, \dots, \gamma_{n-1} + \nu_{n-1}, \nu_n)$$

with the corresponding product of multi-binomial coefficients.

For $\widehat{\Psi}^{-1}$, a sub-multi-index $\nu \leq \alpha_n$ is moved across the final $\mathcal{A}'$ -letter and merged with the terminal block of a parent term. The remaining multi-index $\alpha_n - \nu$ stays in the new terminal block. Empty tuples in the following algorithm are omitted.

Algorithm 7 Sparse matrix for $\widehat{\Psi}$

Require: a bidegree (n, m) and the matrices $\widehat{C}_{n-1, r}^+$ for $0 \leq r \leq m$

Ensure: $\widehat{C}_{n, m}^+$

```

1: if  $n = 0$  then
2:    $\widehat{C}_{0, m}^+ \leftarrow I_{|\text{MPI}(0, m)|}$ 
3: else
4:   initialize  $\widehat{C}_{n, m}^+$  as a zero sparse matrix
5:   for all  $\alpha = (\alpha_0, \dots, \alpha_n) \in \text{MPI}(n, m)$  with increasing rank  $k$  do
6:      $\varrho \leftarrow |\alpha_n|$ 
7:      $\alpha^- \leftarrow (\alpha_0, \dots, \alpha_{n-1})$ 
8:      $k^- \leftarrow \text{rk}_{\text{sym}}(\alpha^-)$ 
9:     for all  $\ell$  such that  $\widehat{C}_{n-1, m-\varrho}^+(k^-, \ell) \neq 0$  do
10:       $(\gamma_0, \dots, \gamma_{n-1}) \leftarrow \text{rk}_{\text{sym}}^{-1}(\ell)$ 
11:      for all componentwise decompositions  $\alpha_n = \nu_0 + \dots + \nu_n$  do
12:         $\delta \leftarrow (\gamma_0 + \nu_0, \dots, \gamma_{n-1} + \nu_{n-1}, \nu_n)$ 
13:         $h \leftarrow \text{rk}_{\text{sym}}(\delta)$ 
14:         $\lambda \leftarrow \prod_{t=0}^{n-1} \prod_{i \in \mathcal{A}''} \binom{\gamma_{t, i} + \nu_{t, i}}{\gamma_{t, i}}$ 
15:         $\widehat{C}_{n, m}^+(k, h) \leftarrow \widehat{C}_{n, m}^+(k, h) + \lambda \widehat{C}_{n-1, m-\varrho}^+(k^-, \ell)$ 
16:      end for
17:    end for
18:  end for
19: end if
20: return  $\widehat{C}_{n, m}^+$ 

```

Proof of Lemma 4.6. We proceed by induction on n . For $n = 0$, both transformations are the identity on the partially symmetrized words over $\mathcal{A}''$, which gives the stated initialization.

Let $n \geq 1$ and consider the output normal form

$$\alpha_0 i_1 \cdots \alpha_{n-1} i_n \alpha_n.$$

Its parent row is indexed by $\alpha^- = (\alpha_0, \dots, \alpha_{n-1})$.

For $\widehat{\Psi}$, the quotient of (6) first appends the final $\mathcal{A}'$ -letter to every term of the parent row and then shuffles the terminal block α_n with the resulting word. A term whose blocks are $\gamma_0, \dots, \gamma_{n-1}, 0$ contributes to the class

$$(\gamma_0 + \nu_0, \dots, \gamma_{n-1} + \nu_{n-1}, \nu_n)$$

for every componentwise decomposition $\alpha_n = \nu_0 + \dots + \nu_n$. The quotient shuffle contributes

$$\prod_{t=0}^{n-1} \prod_{i \in \mathcal{A}''} \binom{\gamma_{t, i} + \nu_{t, i}}{\gamma_{t, i}}.$$

Consequently, Algorithm 7 enumerates precisely the terms of the row recursion. Distinct parent terms or decompositions may yield the same input class, and the corresponding coefficients are correctly accumulated in the same matrix entry.

For $\widehat{\Psi}^{-1}$, the quotient of (7) is obtained by choosing a sub-multi-index $\nu \leq \alpha_n$, merging ν with the terminal block γ_{n-1} of a parent term, appending the final $\mathcal{A}'$ -letter, and retaining $\alpha_n - \nu$ as the new terminal block. The resulting class is

$$(\gamma_0, \dots, \gamma_{n-2}, \gamma_{n-1} + \nu, \alpha_n - \nu),$$

Algorithm 8 Sparse matrix for $\widehat{\Psi}^{-1}$

Require: a bidegree (n, m) and the matrices $\widehat{C}_{n-1, r}^-$ for $0 \leq r \leq m$

Ensure: $\widehat{C}_{n, m}^-$

```

1: if  $n = 0$  then
2:    $\widehat{C}_{0, m}^- \leftarrow I_{|\text{MPI}(0, m)|}$ 
3: else
4:   initialize  $\widehat{C}_{n, m}^-$  as a zero sparse matrix
5:   for all  $\alpha = (\alpha_0, \dots, \alpha_n) \in \text{MPI}(n, m)$  with increasing rank  $k$  do
6:      $\varrho \leftarrow |\alpha_n|$ 
7:      $\alpha^- \leftarrow (\alpha_0, \dots, \alpha_{n-1})$ 
8:      $k^- \leftarrow \text{rk}_{\text{sym}}(\alpha^-)$ 
9:     for all  $\ell$  such that  $\widehat{C}_{n-1, m-\varrho}^-(k^-, \ell) \neq 0$  do
10:       $(\gamma_0, \dots, \gamma_{n-1}) \leftarrow \text{rk}_{\text{sym}}^{-1}(\ell)$ 
11:      for all multi-indices  $\nu \leq \alpha_n$  do
12:         $\delta \leftarrow (\gamma_0, \dots, \gamma_{n-2}, \gamma_{n-1} + \nu, \alpha_n - \nu)$ 
13:         $h \leftarrow \text{rk}_{\text{sym}}(\delta)$ 
14:         $\lambda \leftarrow \prod_{i \in \mathcal{A}''} \binom{\gamma_{n-1, i} + \nu_i}{\gamma_{n-1, i}}$ 
15:         $\widehat{C}_{n, m}^-(k, h) \leftarrow \widehat{C}_{n, m}^-(k, h) + \lambda \widehat{C}_{n-1, m-\varrho}^-(k^-, \ell)$ 
16:      end for
17:    end for
18:  end for
19: end if
20: return  $\widehat{C}_{n, m}^-$ 

```

and the quotient product contributes

$$\prod_{i \in \mathcal{A}''} \binom{\gamma_{n-1, i} + \nu_i}{\gamma_{n-1, i}}.$$

This is exactly the unsigned coefficient accumulated by [Algorithm 8](#).

It remains to identify the sign. At the final recursive step, moving ν into the block preceding the last $\mathcal{A}'$ -letter shifts the position of that letter by $|\nu|$. Hence the sign introduced at this step is $(-1)^{|\nu|}$. Combining this with the induction hypothesis gives

$$(-1)^{\sum_{s=1}^n (g_s - p_s)} = \widehat{\varepsilon}_{n, m}(k, \ell),$$

where $P(\alpha) = \{p_1 < \dots < p_n\}$ and the input multiset placement of rank ℓ induces $\{g_1 < \dots < g_n\}$. Thus all contributions to the same matrix entry have the same sign, which is recovered from the two ranks, while $\widehat{C}_{n, m}^-$ stores only their positive magnitude. This proves both formulas. $\square$

Implementation

The recursive loops in [Algorithms 7](#) and [8](#) are precomputation only. Several contributions may produce the same matrix entry, so their coefficients are first combined. For each bidegree, retain the resulting output-rank list K , input-rank list L , and coefficient list Λ . For a small block for which it is memory-competitive, one may additionally materialize the corresponding dense rank matrix C .

At runtime, flatten the dense $\mathcal{A}'$ -axes into one trailing axis and apply


```

1:  $\mathbf{X} \leftarrow \text{reshape}(\hat{\mathbf{a}}^{(n,m)})$ 
2: if applying  $\hat{\Psi}^{-1}$  then
3:    $\mathbf{X} \leftarrow \hat{\eta}_{n,m} \mathbf{X}$ 
4: end if
5: if a dense rank matrix  $C$  is retained then
6:    $\mathbf{Y} \leftarrow C\mathbf{X}$ 
7: else
8:    $\mathbf{V} \leftarrow \Lambda \mathbf{X}[L]$ 
9:    $\mathbf{Y} \leftarrow \text{scatteradd}(0, K, \mathbf{V})$ 
10: end if
11: if applying  $\hat{\Psi}^{-1}$  then
12:    $\mathbf{Y} \leftarrow \hat{\eta}_{n,m} \mathbf{Y}$ 
13: end if
14: reshape  $\mathbf{Y}$  to the output block shape

```

Here $C = \hat{C}_{n,m}^+$ for the forward map and $C = \hat{C}_{n,m}^-$ for the inverse map. Hence there is no runtime loop over multiset-placement ranks, matrix entries, or dense coordinates. Dense blocks use one rank-axis matrix multiplication; all other blocks use a gather, coefficient scaling, and scatter-add from the retained rank-pair lists. No dense-axis permutation is required.

Finally, consider a nonzero entry with output multiset placement α of rank k and input multiset placement δ of rank h , and let $(a_1, \dots, a_K)$ and $(b_1, \dots, b_K)$ be their blockwise flattenings. The recursive constructions show inductively that $\sum_{l=1}^r b_l \geq \sum_{l=1}^r a_l$ for every $1 \leq r < K$: each step moves multiplicity from a later block to an earlier block, or leaves it in place. It follows from (1) that $h \geq k$. Equality forces the two placements to agree and all distributed or pulled-back multi-indices to vanish. Hence both matrices are upper triangular in the multiset-placement ordering, with diagonal entries equal to one.

Analogously, transposing the precomputed block maps for $\hat{\Psi}$ gives $\hat{\Psi}^*$. For a functional ℓ in partially symmetrized shear coordinates, the shear pairing is evaluated as $\langle \ell, \hat{\mathbf{a}} \rangle_\Psi = \langle \hat{\Psi}^*(\ell), \hat{\mathbf{a}} \rangle$ directly in the partially symmetrized layout. If $\hat{\mathbf{a}} = \hat{\cdot}(\mathbf{a})$, the same pairing may be evaluated against the ordered tensor $\mathbf{a}$ in standard coordinates as $\langle \ell, \hat{\mathbf{a}} \rangle_\Psi = \langle \hat{\cdot}^*(\hat{\Psi}^*(\ell)), \mathbf{a} \rangle$. Thus the partially symmetrized pairing uses no additional class-size weights, while the ordered pairing uses the adjoint lift $\hat{\cdot}^*$.

4.2.4. *Computing signatures.* Since $\hat{\Psi}$ preserves bidegrees,

$$\pi^{N,M} \left(\hat{\Psi}(\widehat{\text{Sig}}(Y)_{s,t}) \right) = \hat{\Psi} \left(\pi^{N,M} \widehat{\text{Sig}}(Y)_{s,t} \right).$$

Thus, one may compute the partially symmetrized bidegree-truncated signature by Algorithm 2 with the partially symmetrized block update, and apply $\hat{\Psi}$ from Section 4.2.3 once at the end. This is the natural procedure when only the final partially symmetrized shear coordinates of the signature are required.

There is also a direct recursion. Since $\hat{\Psi}$ is the identity on the first level, for $z = (z', z'') \in \mathbb{R}^d$ we define

$$\exp_{\hat{\odot}}(z) := \hat{\Psi}(\exp_{\otimes}(z)).$$

The required Horner step only involves the specialized first-level action $\hat{\mathbf{a}} \hat{\odot} z$.

Lemma 4.7. *Let $\hat{\mathbf{a}} \in \hat{T}^{N,M}(\mathbb{R}^d)$, $z = (z', z'') \in \mathbb{R}^d$, and*

$$\hat{\mathbf{c}} := \pi^{N,M}(\hat{\mathbf{a}} \hat{\odot} z).$$

Fix $(n, m) \in \{0, \dots, N\} \times \{0, \dots, M\}$ and $0 \leq k < |\text{MPI}(n, m)|$, and write

$$(\alpha_0, \dots, \alpha_n) = \text{rk}_{\text{sym}}^{-1}(k).$$

Then

$$\hat{\mathbf{c}}_k^{(n,m)} = \begin{cases} \sum_{\substack{i \in \mathcal{A}'' \\ \alpha_{n,i} > 0}} z''_i \hat{\mathbf{a}}_{\text{rk}_{\text{sym}}(\alpha_0, \dots, \alpha_n - \delta_i)}^{(n,m-1)}, & m > 0, \\ 0, & m = 0, \end{cases} + \begin{cases} \left(\prod_{i \in \mathcal{A}''} \binom{\alpha_{n-1,i} + \alpha_{n,i}}{\alpha_{n-1,i}} \right) \hat{\mathbf{a}}_{\text{rk}_{\text{sym}}(\alpha_0, \dots, \alpha_{n-2}, \alpha_{n-1} + \alpha_n)}^{(n-1,m)} \otimes z', & n > 0, \\ 0, & n = 0. \end{cases}$$

For $n = 1$, the initial tuple $(\alpha_0, \dots, \alpha_{n-2})$ in the second term is omitted.

Proof. The quotient map intertwines $\odot$ and $\hat{\odot}$, so the result follows by partially symmetrizing the first-level action from [Lemma 4.4](#).

For the z'' -contribution, appending $i \in \mathcal{A}''$ increases the terminal multi-index by δ_i . Conversely, an output class with $\alpha_{n,i} > 0$ is obtained from the class with terminal multi-index $\alpha_n - \delta_i$. Appending i gives a bijection between the ordered representatives of the latter class and the ordered representatives of the output class ending in i , so no additional coefficient appears.

For the z' -contribution, removing the final $\mathcal{A}'$ -letter merges the two adjacent $\mathcal{A}''$ -blocks. In ordered coordinates these blocks are combined by their ordinary shuffle. After partial symmetrization, all shuffle terms belong to the class with multi-index $\alpha_{n-1} + \alpha_n$, and their multiplicity is

$$\prod_{i \in \mathcal{A}''} \binom{\alpha_{n-1,i} + \alpha_{n,i}}{\alpha_{n-1,i}}.$$

The remaining blocks are unchanged, and tensoring with z' appends the final dense $\mathcal{A}'$ -axis. This gives the second term. $\square$

The bidegree-truncated product

$$\hat{\mathbf{b}}_0 = \pi^{N,M} (\hat{\mathbf{a}} \hat{\odot} \exp_{\hat{\odot}}(z))$$

is therefore computed by the outer recursion of [Algorithm 2](#), using the first-level action from [Lemma 4.7](#).

Implementation

The runtime loops over q and the active bidegrees remain. The z'' -contribution uses the same joint edge axis and the same precomputed direct or destination-ordered plan as in the standard partially symmetrized recursion. Thus all $\mathcal{A}''$ -letters and input ranks are processed together. For the z' -contribution, precompute the input-rank and coefficient arrays $L_{n,m}$ and $\Lambda_{n,m}$. The fixed-bidegree update is

- 1: $\hat{\mathbf{b}}_{q-1}^{(n,m)} \leftarrow \hat{\mathbf{a}}^{(n,m)}$
- 2: **if** $m > 0$ **then**
- 3: form all values $q^{-1} z''_i (\hat{\mathbf{b}}_q^{(n,m-1)})_k$ on the joint edge axis (i, k)
- 4: add them by the retained direct or destination-ordered plan
- 5: **end if**
- 6: **if** $n > 0$ **then**
- 7: $\mathbf{V} \leftarrow \hat{\mathbf{b}}_q^{(n-1,m)}[L_{n,m}]$
- 8: $\hat{\mathbf{b}}_{q-1}^{(n,m)} += q^{-1} \Lambda_{n,m} \mathbf{V} \otimes z'$
- 9: **end if**

There is no runtime loop over letters or multiset-placement ranks. The z' -term is one dense gather, scaling, tensor product, and direct addition in output-rank order; no reduction or dense-axis permutation is required. Only two Horner states are live.

In the z' -contribution, the entire shuffle of the two terminal $\mathcal{A}''$ -blocks collapses to one source rank and one multi-binomial coefficient.

5. PARALLEL SIGNATURE COMPUTATION

The Horner schemes above update complete bidegree blocks through vectorized array operations. A complementary approach proposed in [4] computes signature coefficients directly in the word basis. For each target word, one computational unit evolves the signature coefficients of its prefix chain through all path increments. The time iteration within one unit remains sequential, whereas the units associated with distinct target words are independent. We explain how this word-parallel approach interfaces with the ordered and partially symmetrized bidegree layouts.

5.1. Wordwise recursion. For reference, we first recall the wordwise signature recursion from [4, Secs. 3.1–3.2]. The construction in this subsection is not new; we repeat it only to fix notation and to make the subsequent adaptation to the ordered and partially symmetrized bidegree layouts self-contained.

Let Y be piecewise linear on $s = t_0 < \dots < t_J = t$, with increments $\Delta Y_j = Y_{t_{j-1}, t_j}$, and set $\mathbf{S}_j := \text{Sig}(Y)_{s, t_j}$. For $w = i_1 \dots i_r \in \mathcal{W}_{\mathcal{A}}$, write $w_{[q]} := i_1 \dots i_q$ and $w_{[0]} := \emptyset$. Chen’s identity gives

$$\langle w_{[p]}, \mathbf{S}_j \rangle = \sum_{q=0}^p \langle w_{[q]}, \mathbf{S}_{j-1} \rangle \frac{\prod_{u=q+1}^p \Delta Y_j^{i_u}}{(p-q)!}, \quad p = 0, \dots, r,$$

with the empty product equal to one. The following in-place Horner scheme computes the terminal coefficient while retaining only the coefficients of the prefixes of w .

Algorithm 9 Signature computation over one prefix chain

Require: a word $w = i_1 \dots i_r$ and increments $\Delta Y_1, \dots, \Delta Y_J$

Ensure: $\langle w, \text{Sig}(Y)_{s, t} \rangle$

```

1:  $a_0 \leftarrow 1$  and  $a_p \leftarrow 0$  for  $p = 1, \dots, r$ 
2: for  $j = 1, \dots, J$  do
3:   for  $p = r, \dots, 1$  do
4:      $h \leftarrow 0$ 
5:     for  $q = 0, \dots, p-1$  do
6:        $h \leftarrow \frac{\Delta Y_j^{i_{q+1}}}{p-q} (a_q + h)$ 
7:     end for
8:      $a_p \leftarrow a_p + h$ 
9:   end for
10: end for
11: return  $a_r$ 
```

The descending loop over p ensures that every a_q used in the update still equals $\langle w_{[q]}, \mathbf{S}_{j-1} \rangle$. Thus one computational unit requires $r+1$ local scalar values and performs $\mathcal{O}(Jr^2)$ arithmetic operations. Assigning one prefix chain to each target word repeats the computation of shared prefixes, but removes any dependence between distinct computational units; this is the granularity used in [4].

5.2. Ordered bidegree layout. The set of words retained by bidegree truncation is

$$\mathcal{W}_{\mathcal{A}}^{N,M} := \{w \in \mathcal{W}_{\mathcal{A}} : |w|_{\mathcal{A}'} \leq N, \quad |w|_{\mathcal{A}''} \leq M\}.$$

It is prefix-closed, since taking a prefix cannot increase either component of the bidegree. Hence [Algorithm 9](#) applies directly to every coordinate of the bidegree-truncated signature.

For fixed (n, m) , interleaving the two letter sequences according to the placement gives a bijection

$$\text{Pl}(n, m) \times (\mathcal{A}')^n \times (\mathcal{A}'')^m \longrightarrow \{w \in \mathcal{W}_{\mathcal{A}} : |w|_{\mathcal{A}'} = n, \quad |w|_{\mathcal{A}''} = m\}.$$

Replacing the placement by its colexicographic rank is exactly the block layout in (2). Thus the computational units for bidegree (n, m) may be indexed directly by the scalar entries of $\mathbf{S}^{(n,m)}$, without first enumerating the words in a separate ordering.

Implementation

For a fixed bidegree (n, m) , launch one computational unit for every scalar entry of the block, hence

$$|\text{Pl}(n, m)| |\mathcal{A}'|^n |\mathcal{A}''|^m$$

units in total. The ordinary flattening of the block axes gives the unit index. The placement rank is decoded through the precomputed inverse-rank table, and the two dense letter tuples are interleaved according to that placement. The unit then applies [Algorithm 9](#) and writes its terminal value directly to the same array entry. The prefix values remain local to the unit, so no ranks of the prefixes are required. There is no output permutation or rank-axis accumulation. The loops over the path increments and the prefixes remain inside each unit, while all scalar entries of a fixed bidegree block are independent. Processing one bidegree at a time gives the same word length and loop bounds to all units in the corresponding launch.

5.3. Partially symmetrized bidegree layout. The direct reuse of the ordered word-parallel computation is to evaluate all ordered representatives and then apply the partial symmetrization map. Since one partially symmetrized coordinate is the sum of the corresponding ordered coordinates, this reduction may be fused with the final writes by mapping each ordered word to its multiset-placement rank. This avoids storing the complete ordered output, but it does not reduce the wordwise arithmetic and requires accumulation when several representatives have the same target rank.

A quotient-native alternative follows from the partially symmetrized first-level update. Let

$$[w] = \alpha_0 i_1 \alpha_1 \cdots i_n \alpha_n$$

be a quotient word in normal form. Its multiset-prefix set is

$$\begin{aligned} \text{Pref}_{\text{sym}}([w]) &:= \{\beta : 0 \leq \beta \leq \alpha_0\} \\ &\cup \bigcup_{p=1}^n \{\alpha_0 i_1 \cdots \alpha_{p-1} i_p \beta : 0 \leq \beta \leq \alpha_p\}, \end{aligned}$$

where the inequalities are componentwise. These are precisely the quotient classes containing a prefix of an ordered representative of $[w]$. The cardinality of this set is

$$|\text{Pref}_{\text{sym}}([w])| = \sum_{p=0}^n \prod_{i \in \mathcal{A}''} (\alpha_{p,i} + 1).$$

Lemma 5.1. *The set $\text{Pref}_{\text{sym}}([w])$ is closed under the predecessor maps in the partially symmetrized first-level action. Consequently, the coordinate $\langle [w], \widehat{\text{Sig}}(Y)_{s,t} \rangle$ can be computed independently by restricting the partially symmetrized product–Horner recursion to this set.*

Proof. Consider an element whose terminal multi-index is β . Its $\mathcal{A}''$ -contribution only uses the elements obtained by replacing β by $\beta - \delta_i$ for those $i \in \mathcal{A}''$ with $\beta_i > 0$. These elements belong to the same componentwise interval. If $\beta = 0$ and the element ends in $i_p \in \mathcal{A}'$, its $\mathcal{A}'$ -contribution uses the preceding element ending in the complete block α_{p-1} . This element also belongs to $\text{Pref}_{\text{sym}}([w])$. Hence the set is closed under all predecessor operations in the partially symmetrized first-level action. The product–Horner recursion consists only of repeated applications of this action, so no coordinate outside the set is required. $\square$

Implementation

For $k = \text{rk}_{\text{sym}}(\alpha_0, \dots, \alpha_n)$, precompute the elements of $\text{Pref}_{\text{sym}}([w])$ and their predecessor edges. The combinatorial graph depends only on the multiset-placement rank k ; its p -th $\mathcal{A}'$ -edge reads the increment coordinate selected by the p -th dense $\mathcal{A}'$ -index. One computational unit may then be assigned to every scalar entry

$$(k, i_1, \dots, i_n)$$

of the block in (3). It keeps the values on the multiset-prefix set locally, applies the partially symmetrized Horner recursion through the path increments, and writes only the coordinate of $[w]$. Distinct output entries are independent, and no ordered $\mathcal{A}''$ -axes or rank-axis reduction are required. The path increments and the local Horner depth remain sequential within the unit. The local coordinate set has cardinality $|\text{Pref}_{\text{sym}}([w])|$, so its working memory varies with the multiset-placement rank. When this local problem is too large for one computational unit, the ordered word-parallel computation followed by reduction onto the multiset-placement layout gives the direct fallback.

The ordered reduction is an immediate implementation of partial symmetrization. The quotient-native construction in Lemma 5.1 avoids the expansion into ordered representatives, but its efficiency depends on the sizes of the multiset-prefix sets and on the available local memory.

5.4. Shear coordinates. The word-parallel recursion is naturally formulated in standard coordinates. Its ordered output already has the layout required by Ψ , while the quotient-native output has the layout required by $\widehat{\Psi}$. The corresponding shear-coordinate signatures are therefore obtained by

$$\Psi(\pi^{N,M} \text{Sig}(Y)_{s,t}) \quad \text{and} \quad \widehat{\Psi}(\pi^{N,M} \widehat{\text{Sig}}(Y)_{s,t}),$$

using the transformations constructed above.

We do not propose a direct word-parallel recursion in shear coordinates. In the ordered case, right multiplication by an $\mathcal{A}'$ -letter joins two adjacent $\mathcal{A}''$ -blocks through an ordinary shuffle; after partial symmetrization, the same action contributes a multi-binomial coefficient. A shear-coordinate output is therefore not closed on an ordinary prefix chain. The direct shear-coordinate Horner schemes of Lemmas 4.4 and 4.7 remain the appropriate alternatives when the intermediate states are required in shear coordinates.

REFERENCES

- [1] Aviezri S. Fraenkel and Moshe Mor. Combinatorial compression and partitioning of large dictionaries. *The Computer Journal*, 26(4):336–343, 1983.
- [2] Paul P. Hager and Luca Pelizzari. Expected signatures via partial integration, coordinate change and symmetrization. arXiv:2607.29534 [math.PR], 2026.
- [3] Donald E. Knuth. *The Art of Computer Programming, Volume 4A: Combinatorial Algorithms, Part 1*. Addison-Wesley, 2011.
- [4] Tobias Nygaard. pathsig: A GPU-accelerated library for truncated and projected path signatures, 2026.